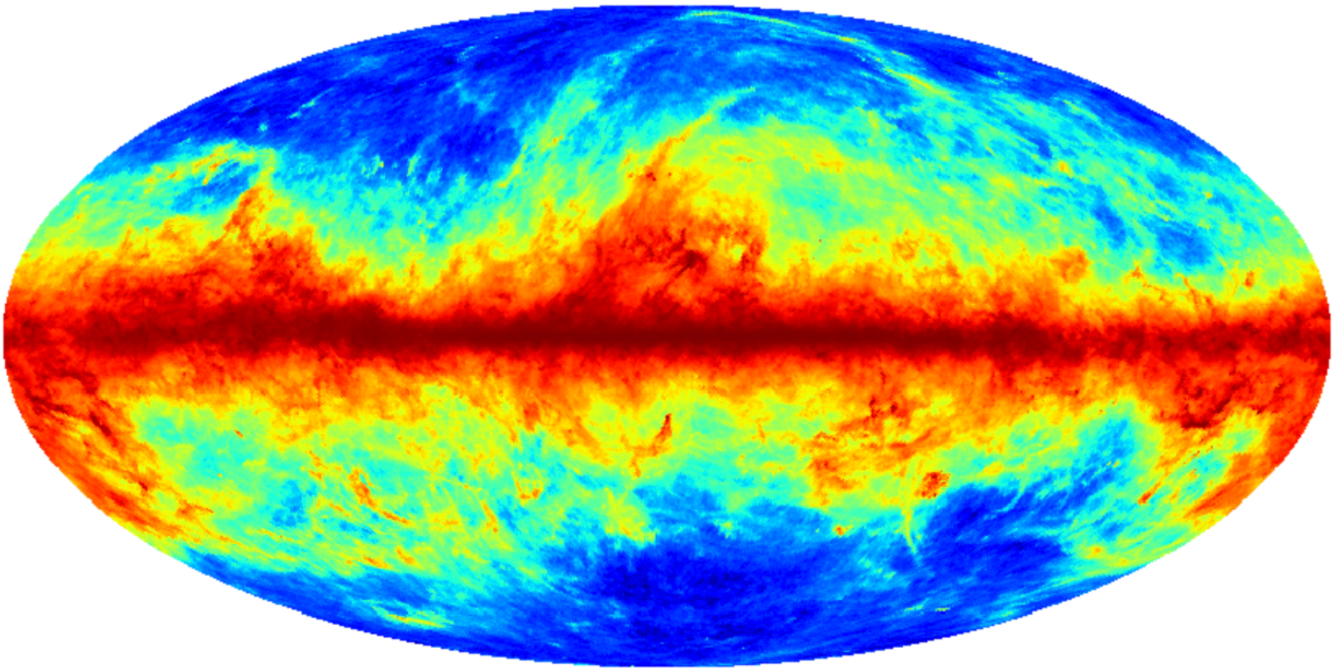
M2 INTERNSHIP REPORT

Characterization of the spectral complexity of galactic foregrounds for the study of B modes of the cosmic microwave background



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1 Remerciements

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Part I

Introduction

The Cosmic Microwave Background (CMB) is the residual electromagnetic radiation from the Big Bang, observable in all directions of the Universe. It carries an image of the Universe 380,000 years after the Big Bang, a period when electrons combined with protons to form neutral atoms, allowing light to move freely.

The small temperature and polarization variations in the CMB, known as anisotropies, provide information about the conditions of the early Universe. However, studying these anisotropies is complex, particularly due to the presence of foreground emissions that must be carefully separated from the primordial signal.

In this report, we focus on the polarization anisotropies of the CMB, specifically on polarization patterns known as B-modes. According to inflation theory, the Universe underwent an exponentially accelerated expansion just after the Big Bang, generating primordial gravitational waves. These primordial gravitational waves would have left a characteristic imprint on the CMB in the form of a very faint signal, observable as B-modes of polarization.

The amplitude of these B-modes, quantified by the parameter r, is related to the energy scale of inflation. Observing these B-modes is thus a challenge for cosmology because detecting them would be direct evidence of an inflationary phase and would provide a better understanding of its energy scale, enabling the testing of fundamental physics at scales that will never be accessible on Earth.

The LiteBIRD satellite mission, scheduled for launch in 2032, aims to measure r with unprecedented precision. LiteBIRD, a mission of the Japan Aerospace Exploration Agency (JAXA), has the primary objective of detecting the CMB B-modes while minimizing contamination from galactic foreground emissions. To achieve this goal, it is crucial to perform a very fine component separation to distinguish the primordial signals from the contamination of galactic foreground emissions. Among these contaminations, synchrotron emission and dust emission play a major role, as both are significantly polarized in the frequency range of CMB studies. During this internship and in this report, I focused on the precise characterization of synchrotron and dust emissions to study the CMB. In particular, I worked on accounting for their spectral complexity due to mixing along the lines of sight or within the instrumental beams, of regions of the interstellar medium with different physical conditions.

This report will be divided into two parts. In the first part, we will focus on the study of CMB B-modes in general, and in the second part, we will explore an innovative method for component separation in CMB maps contaminated by foreground emissions. We will present the principles of this method, the results of simulations, and the prospects within the framework of the LiteBIRD mission.

Part II

CMB B-modes

2 The Cosmic Microwave Background (CMB)

The discovery of the Cosmic Microwave Background in 1965 by Arno Penzias and Robert Wilson [1] played a fundamental role in confirming the Big Bang model. Penzias and Wilson observed a constant background signal persisting in all directions of space after eliminating all potential sources of noise (including pigeon droppings that occupied the telescope).

Researchers affiliated with Princeton University, led by Robert Dicke, realized that this matched the prediction of the Cosmic Microwave Background, a residual signature of the Big Bang [2]. This historic discovery not only confirmed the validity of the Big Bang cosmological model but also paved the way for a deeper exploration of the fundamental physics of the early Universe.

2.1 Recombination and the Surface of Last Scattering

After the Big Bang, the Universe was characterized by extremely high density and temperature, resulting in the complete ionization of the Universe. The primordial photons underwent Thomson scattering through the free electrons in the plasma that filled the Universe at that time.

As the Universe expanded, the density and temperature gradually decreased. About 380,000 years after the Big Bang, conditions became favorable for the recombination of electrons and nuclei: Electrons combined with nuclei to form the first atoms. As photon scattering on free electrons became increasingly rare, the mean free path of photons increased significantly until it became comparable to the size of the Universe. This phenomenon is known as photon decoupling.

The Cosmic Microwave Background (CMB) we observe today consists of photons after a final Thomson scattering, marking the formation of the surface of last scattering.

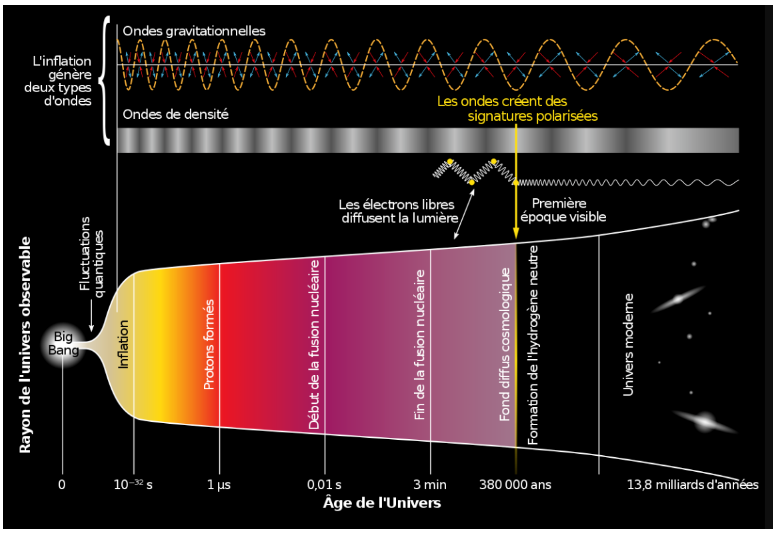


Figure 1: Diagram representing the history of the Universe in a Λ-CDM model, credit: BICEP/Keck Collaboration

Due to the thermodynamic equilibrium between photons, electrons, and protons during recombination, the intensity of the CMB's electromagnetic spectrum follows Planck's law, where $T_{\rm CMB}$ represents the temperature of the CMB ($T_{\rm CMB} = 2.72548 \pm 0.00057$ K) [3]:

$$I(\nu, T_{\rm CMB}) = \frac{2h\nu^3}{c^2} \left(\exp\left(\frac{h\nu}{kT_{\rm CMB}} \right) - 1 \right)^{-1}. \tag{1}$$

For a detailed study of the theoretical context and the problems solved by modern cosmological models (notably concerning the issues related to the Big Bang model, which an inflationary model resolves), we refer to Appendices A and B.

One of the major issues with the hot Big Bang model without inflation is the extreme homogeneity of the CMB temperature, with fluctuations on the order of a few 10^{-5} K. We map these temperature and polarization anisotropies, which represent cosmological information.

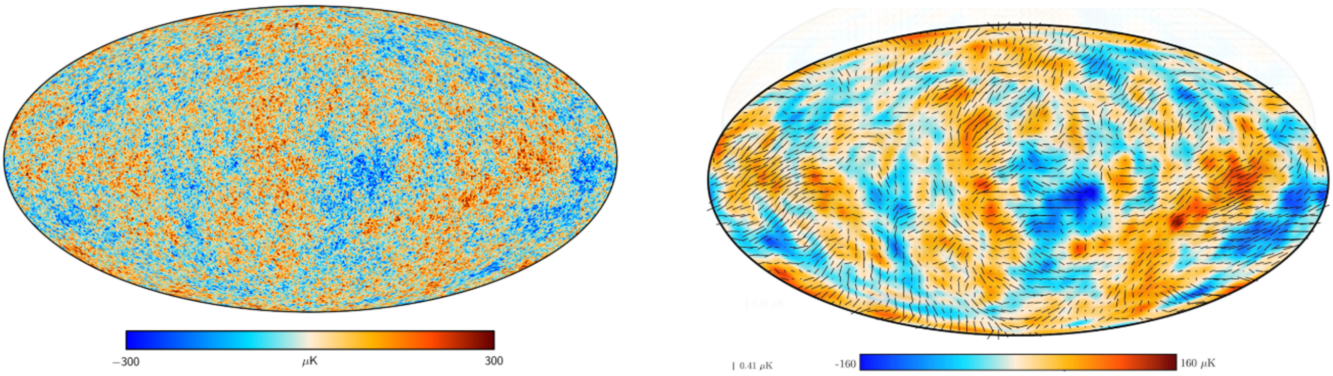


Figure 2: Mollweide projection maps of CMB anisotropies in intensity (left) and polarization (right) [4]

2.2 CMB Anisotropies

1. **The Cosmological Dipole**:

   The most significant anisotropy observed in the CMB is due to our relative motion with respect to it. This relative motion is divided into two distinct components. The first is attributed to our movement within the solar system, known as the orbital dipole. The second is the movement of our solar system relative to the cosmic reference frame, called the solar dipole.

   These various motions induce a shift in CMB photons due to the relativistic Doppler effect. This shift, resulting from the variation in the recession velocity of photons depending on our relative motion, translates into an observed anisotropy in the CMB intensity in different directions of the sky. We are moving at a velocity of 369.82 ± 0.11 km s^{-1} towards the constellation Leo. The amplitude of the anisotropies is measured at $(3362.08 \pm 0.99) \times 10^{-3}\ \mu$K [4].

2. **Primordial Temperature Anisotropies**:

   The primordial temperature anisotropies are the anisotropies imprinted on the CMB at the time of the formation of the surface of last scattering. The deviations from the mean CMB temperature are mainly due to the following three processes:

   (a) Density perturbations: The anisotropies generated by these perturbations are proportional to the density variations predicted by the inflation model (see Appendix B for more details on inflation).

   (b) Doppler perturbations: In the primordial fluid, density perturbations move relative to each other, inducing temperature variations through the Doppler effect proportional to the velocity of the density fluctuations.

   (c) Gravitational perturbations: also known as the Sachs-Wolfe effect: When photons travel through a perturbed gravitational field, they gain or lose energy, which results in a shift in their wavelength. Temperature anisotropies are then proportional to the gravitational perturbations.

3. **Polarization Anisotropies**:

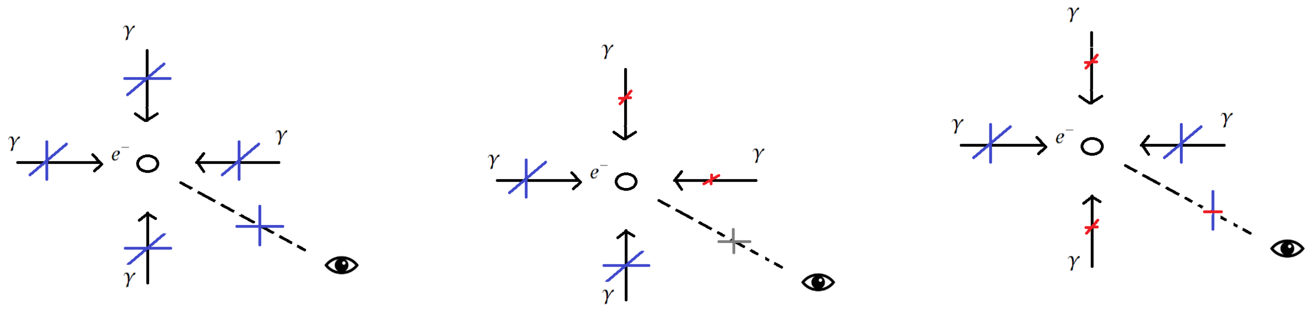
   Before decoupling, photons interacted with electrons through Thomson scattering, as described by:

$$e^{-} + \gamma \longrightarrow e^{-} + \gamma \tag{2}$$

Consider a stationary electron undergoing scattering with a photon. Denoting ϵ and ϵ' as the unit polarization vectors of the incident and scattered flux, respectively, the differential cross-section for the scattering can be expressed as:

$$\frac{d\sigma}{d\Omega} = \frac{3\sigma_{\mathrm{T}}}{8\pi}|\epsilon \cdot \epsilon'|^2. \tag{3}$$

As illustrated in Figure 3 and indicated by Equation (3), only quadrupolar anisotropies of the incident intensity can generate non-zero polarization. Indeed, in Figure 3, it is observed that the superposition of polarities for a monopole or dipole distribution leads to the cancellation of polarization components, resulting in zero polarization for the observer. However, for a quadrupole, the vertical and horizontal components do not have the same amplitude, meaning that the resulting polarization does not cancel out.



(a) monopole (b) dipole (c) quadrupole

Figure 3: Comparison between monopoles, dipoles, and quadrupoles of incident intensity during Thomson scattering to generate linear polarization. High-intensity polarization is shown in blue, low-intensity in red, and medium intensity in gray.

We characterize the polarization of an electromagnetic signal by the Stokes parameters: intensity I and polarization parameters Q, U, and V. The first two polarization parameters define linear polarization, while the last defines circular polarization, which is not generated by Thomson scattering and will be neglected in this document.

We can now enumerate the different perturbations that generate quadrupolar anisotropies:

(a) **Scalar Perturbations**: In the vicinity of the under- and over-densities corresponding to primordial scalar perturbations, quadrupolar anisotropies will be generated in the Thomson scattering of the CMB. These perturbations, created during inflation, influence the density distribution of the primordial universe. Thus, when photons interact with free electrons via Thomson scattering, the geometry of this interaction around the regions of over- and under-density produces observable quadrupolar anisotropies in the CMB.

Consider the case of an over-density: by definition, an over-density has a stronger gravitational potential at its center. Thus, electrons near the center of gravity experience a stronger gravitational attraction than those further away. In other words, electrons near the center of gravity acquire higher velocities. In its own reference frame, the electron moves away from the electrons in front and behind it, inducing an interaction with a photon that will be redshifted (due to the Doppler effect).

Conversely, due to their attraction toward the center of the over-density, electrons on the same density isocontour move closer together, resulting in an interaction with a photon that is blueshifted. These perturbations generate a quadrupolar anisotropy, and the photon undergoing this interaction will be polarized. This particular polarization is called E-mode and is shown in Figure 4. The same (inverse) phenomenon occurs during under-density perturbations.

(b) **Tensor Perturbations**:

A second phenomenon capable of generating quadrupolar anisotropies is the propagation of gravitational waves. In the electron's reference frame, the passage of a gravitational wave deforms the surrounding space, thereby generating quadrupolar anisotropies. The resulting photons will be polarized both in E-mode and in another polarization mode called B-mode, depicted in Figure 4.

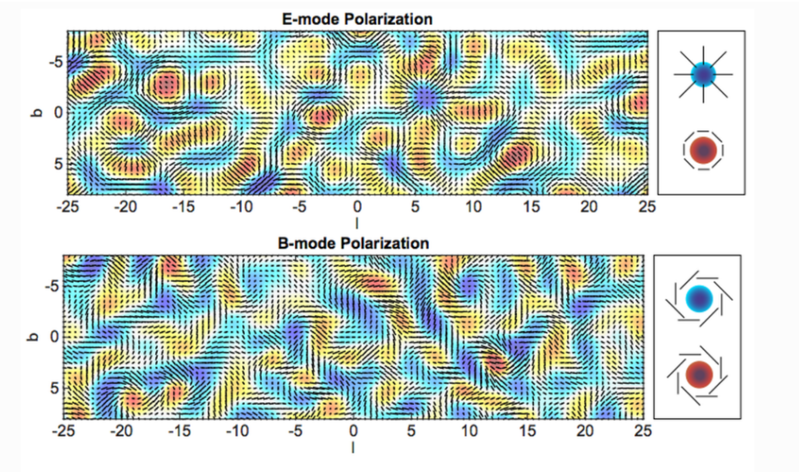


Figure 4: Map of pure E-mode (top) and pure B-mode (bottom) polarizations. The E and B polarization patterns are shown in the panels on the right [5]

For a more detailed analysis of scalar and tensor perturbations, we refer to Appendix C.

4. **Secondary Anisotropies**: Before reaching us, CMB photons undergo various interactions. We will mention here only the most significant one in the context of our study on measuring B-modes.

   Gravitational Lensing Effect: When photons pass through large-scale structures, they are deflected by the intense gravitational fields generated by these structures. This effect mixes the polarization patterns on the sky and creates secondary B-modes from the primordial E-modes.

   Many other effects, such as the Sunyaev-Zel'dovich (SZ) effect, the Integrated Sachs-Wolfe (ISW) effect, or reionization, can also affect CMB photons during their journey from the surface of last scattering to us.

2.3 Statistical Properties of the CMB

Due to the stochastic nature of the primordial perturbations that left their imprint on the CMB, the analysis of anisotropies is conducted statistically. Since the primordial perturbations generated during inflation are predicted with a Gaussian distribution, the same applies to the CMB anisotropies.

2.3.1 Spherical Harmonics

Spherical harmonics are a set of orthogonal functions that allow for the decomposition of any field on the sphere. They are used to decompose the temperature fluctuations of the CMB into different Fourier modes based on their spatial and angular structures.

For temperature, we project the deviations between $T(\theta,\phi)$ and the average temperature $\bar{T}$:

$$\frac{T(\theta,\phi)-\bar{T}}{\bar{T}} = \sum_{\ell=+1}^{+\infty}\sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\theta,\phi), \tag{4}$$

where $Y_{\ell m}$ represents the spherical harmonic functions. The parameter ℓ is the multipole, inversely proportional to the angular scales on the sky, and the $a_{\ell m}$ represent the coefficients associated with the different spherical harmonics.

For polarization, the Stokes parameters Q and U are jointly projected in the space of spin-weighted spherical harmonics:

$$(Q \pm iU)(\vec{n}) = \sum_{\ell,m} a_{\pm 2,\ell m} \cdot_{\pm 2} Y_{\ell m}(\vec{n}). \tag{5}$$

From the spherical harmonic decomposition coefficients $a_{\pm 2\ell m}$, we can construct two observables, E and B [6].

$$\begin{matrix} E(\vec{n}) = \sum_{\ell m} a^E_{\ell m} Y_{\ell m}(\vec{n}) & & a^E_{\ell m} = -\frac{1}{2}(a_{2,\ell m} + a_{-2,\ell m}) \\ & \text{with} & \\ B(\vec{n}) = \sum_{\ell m} a^B_{\ell m} Y_{\ell m}(\vec{n}) & & a^B_{\ell m} = \frac{i}{2}(a_{2,\ell m} - a_{-2,\ell m}) \end{matrix} \tag{6}$$

The parameters Q and U depend on the chosen reference frame. However, by transforming these parameters and working with the E and B parameters, the latter become reference-frame invariant. This transformation allows for easier interpretation. It can thus be shown that scalar perturbations generate only E modes, while tensor perturbations generate both E and B modes. Hence, measuring primordial B modes allows for the study of tensor anisotropies.

2.3.2 Angular Power Spectra

From the three observables T, E, and B constructed in the space of spherical harmonics, we define six angular power spectra:

$$C_\ell^{XX'} = \langle a^X_{\ell m} \cdot a^{X'*}_{\ell m} \rangle, \tag{7}$$

for all $(X, X') \in \{T, E, B\}$. The spectra with $X = X'$ are auto-correlation angular power spectra, while those for $X \neq X'$ are cross-correlation angular power spectra, also known as cross-spectra. The $\langle \dots \rangle$ represents the ensemble average over all possible configurations of CMB anisotropies for a given multipole ℓ. We construct an estimator for C_ℓ, averaged over all possible values of m for a given ℓ:

$$\hat{C}_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{+\ell} |a_{\ell m}|^2. \tag{8}$$

This estimator has an intrinsic variance, called cosmic variance, which is related to the fact that there are not enough independent m modes at large scales (hence, particularly significant for low values of ℓ):

$$\mathrm{Var}(C_\ell^{XX'}) = \frac{2}{2\ell+1} \left(C_\ell^{XX'}\right)^2 \tag{9}$$

Throughout this study, when manipulating the angular power spectra, we will use the quantity[1] $D_\ell^{XX'} := \ell(\ell+1)/2\pi C_\ell^{XX'}$.

The angular power spectra of CMB temperature and polarization anisotropies are shown in Figure 5.

In temperature, we observe three distinct regions:

- At large scales $\ell \lesssim 100$, a plateau, called the Sachs-Wolfe plateau, where only gravitational anisotropies are visible.
- For $\ell \geq 100$, a series of acoustic peaks is observed. These peaks result from the competition between gravitational attraction and radiation pressure, amplifying density fluctuations through processes of compression and rarefaction.

[1]This definition allows for flattening the power spectra, making them easier to read.

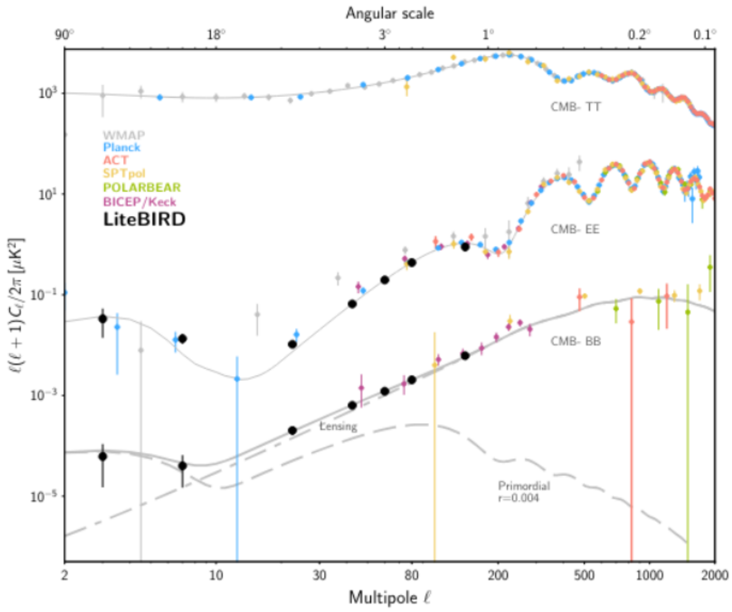


Figure 5: Angular power spectra D_ℓ^{TT}, D_ℓ^{EE}, and D_ℓ^{BB}, in μK^2, as a function of the multipole ℓ. The two components, Lensing and Primordial, are represented by dashed lines. The amplitude of the primordial spectrum is quantified by the parameter $r = 0.004$. The spectra of the concordance model Λ-CDM (solid gray lines) are compared to measurements from various experiments (points, see legend). [7]

- At small scales ($\ell \geq 1000$), we see a damping region. This region mainly results from the dissipation of small-scale density fluctuations due to Thomson scattering of photons on electrons before decoupling. This process attenuates anisotropies at fine angular scales, leading to a decrease in the amplitude of fluctuations in the power spectrum.

In polarization, we observe that the oscillations of the angular power spectrum D_ℓ^{EE} are out of phase with the D_ℓ^{TT} spectrum. This shift arises from the fact that polarization spectra are dominated by the Doppler effect, as discussed in paragraph **??**, unlike the D_ℓ^{TT} temperature spectrum, which is dominated by acoustic oscillations.

For the D_ℓ^{BB} power spectrum, we observe two distinct components in Figure 4, referred to as "lensing" and "primordial." We will study these components separately.

- **Lensing spectrum**: As mentioned earlier in 2, the gravitational lensing effect perturbs polarization patterns. In particular, this will cause a mixing of the D_ℓ^{EE} and D_ℓ^{BB} spectra. Given that the D_ℓ^{EE} power spectrum is several orders of magnitude more intense than D_ℓ^{BB}, it results, to the first order, in a leakage of D_ℓ^{EE} into D_ℓ^{BB}.
- **Primordial spectrum**: As also mentioned earlier in 3b, this spectrum carries the imprint of tensor perturbations generated during the inflationary phase. Its amplitude is directly related to the parameter r, defined as the ratio of tensor to scalar perturbations (For a more detailed study of this parameter, we refer to Appendix C). To date, the scalar-to-tensor ratio has not yet been measured, and only an upper limit has been established.

2.4 Measuring the CMB

An international effort is underway to detect primordial B-modes, as this signal, being a direct probe of the inflationary phase, constitutes one of the major goals of observational cosmology. The most precise limit to date, provided by BICEP/Keck[2] and Planck, is $r \leq 0.032$, 95% C.L. [8]. Future missions such as CMB-S4

[2] BICEP/Keck is located at the South Pole to enable long-duration observations. Given its polar position, the visible sky remains relatively fixed, facilitating continuous observations.

and LiteBIRD aim to constrain the parameter r with a precision[3] down to $r = 0.001$ [10, 7] (this corresponds to a measurement of the CMB temperature with a precision on the order of nano-Kelvin).

The LiteBIRD mission [7], an acronym for "Lite (Light) satellite for the studies of B-mode polarization and Inflation from cosmic Background Radiation Detection," plays a central role in this quest. Scheduled for launch in 2032, LiteBIRD is a mission led by the Japanese space agency (JAXA), with significant involvement from CNES and IRAP. LiteBIRD will be equipped with three instruments—LFT, MFT, and HFT (respectively "Low, Medium, and High Frequency" Telescopes)—covering a wide range of frequencies from 40 GHz to 402 GHz, allowing it to distinguish the cosmological signal from galactic emissions, which we will detail in the following sections. Two of these instruments (LFT and MFT) are under European responsibility, while HFT is under Japanese responsibility.

Such precision requires mastery of systematic and instrumental effects, as well as an extremely precise component separation. This constitutes a real challenge, and the goal of the work presented in this document is aligned with this ambition. LiteBIRD uses superconducting transition edge sensors (TES). A TES detector operates on the principle of maintaining the superconducting material just below the critical temperature. When a photon strikes the detector, it excites the electrons in the material, slightly increasing its temperature and causing it to transition into a normal resistive state. This phase transition causes a change in electrical resistance, which can be measured and interpreted as a temperature variation. LiteBIRD will be equipped with approximately 4,000 TES detectors, distributed across its three instruments, to ensure maximum sensitivity and precision.

3 Foreground Emissions

To explore the cosmic background that is the CMB, it is essential to examine the foregrounds that can obscure our observations of the CMB. The foreground emissions are varied, occurring at different scales, and can be either polarized or unpolarized.

Among the primary sources of contamination of the CMB, dust emission and synchrotron emission are the only ones with significant polarization. We will examine these two emissions in detail in the following subsections.

For other major emissions, we refer to Appendix D for a more detailed study on free-free emission, AME, molecular emission lines, the cosmic infrared background (CIB), point sources, and zodiacal light.

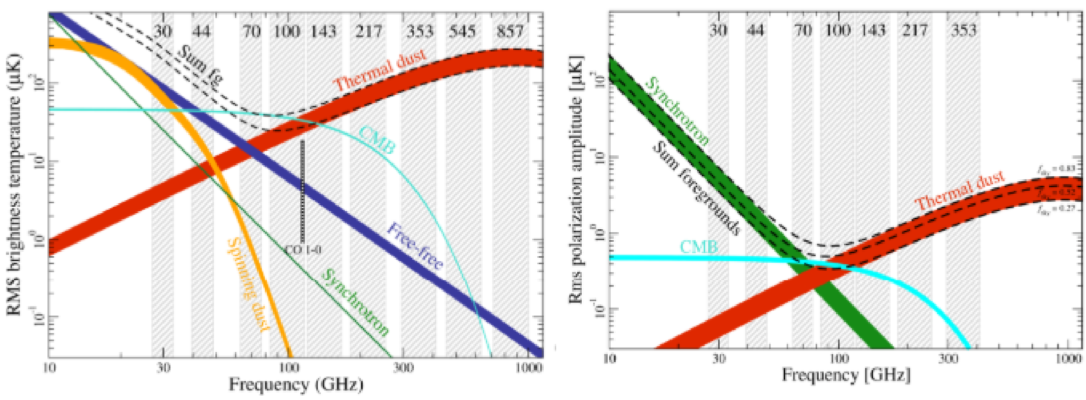


Figure 6: Comparison of the SEDs of different foregrounds. On the left, the SEDs in intensity; on the right, the SEDs in polarization. The gray vertical bands represent the frequency channels of the Planck satellite. The thickness of the curves indicates the difference in amplitude between two fractions of the sky, denoted by f_{sky}. For intensity, the difference is between $f_{\text{sky}} = 0.81$ and $f_{\text{sky}} = 0.93$. For polarization, it is between $f_{\text{sky}} = 0.73$ and $f_{\text{sky}} = 0.93$. [4]

The study of synchrotron and dust polarization is of interest beyond the CMB, as they can trace the structure of the Galactic magnetic field, as well as other phenomena such as the nature of dust grains, their

[3]Many inflationary models favor values of r between 0.1 and 0.001 [9], which could be probed by upcoming missions [10, 7].

alignment, cosmic rays, and more.

3.1 Synchrotron Emission

Synchrotron emission occurs when charged particles, such as electrons, are accelerated in a magnetic field. These particles follow curved trajectories due to the interaction between their electric and magnetic fields, emitting photons as they are deflected. In the interstellar medium (ISM), electrons are accelerated by the Galactic magnetic field and emit in the frequency range observed for the CMB. This synchrotron emission dominates at frequencies $\lesssim 70$ GHz. [for a review see 11].

The spectral energy distribution (SED) of this emission can be expressed, for any pixel in a direction $\vec{n}$ on the sky, by a power law of the observation frequency ν:

$$I_\nu^{\mathrm{s}}(\vec{n}) = A_{\mathrm{s}}(\vec{n}) \left(\frac{\nu}{\nu_0^{\mathrm{s}}}\right)^{\beta_{\mathrm{s}}(\vec{n})}, \tag{10}$$

where $\beta_{\mathrm{s}}(\vec{n})$ represents the spectral index of the synchrotron. This spectral index can change with the direction $\vec{n}$; in this case, $\beta_{\mathrm{s}}(\vec{n})$ is a sky map of spectral indices. $A_{\mathrm{s}}(\vec{n})$ is the intensity of the synchrotron signal at the reference frequency ν_0^{s}. It is important to note that the observed spectral distribution is not perfectly described by a power law [12]. However, this approximation is valid in the frequency range of CMB observations [13]. The polarization of synchrotron emission is perpendicular to the magnetic field and can reach up to 75% of the total intensity of synchrotron emission [14].

A more refined model can account for emissions from regions with complex physical conditions, such as jets emitted by active galactic nuclei (AGN) or matter ejected from neutron stars or black holes, by introducing a frequency dependence of the spectral index: $\beta_{\mathrm{S}}(\vec{n}) \longrightarrow \beta_{\mathrm{S}}(\vec{n}) + c(\vec{n}) \ln\left(\frac{\nu}{\nu_0^{\mathrm{S}}}\right)$, known as the *running* model [15].

3.2 Dust Emission

Dust emission originates from dust grains present in our Galaxy, the Milky Way, as well as in other galaxies.[4] These grains are heated by starlight in the visible spectrum and then emit thermally in the infrared and sub-millimeter wavelengths.

This emission is primarily observed in star-forming regions, dense molecular clouds, or the spiral arms of galaxies, but it also has a diffuse component present in regions with lower column density, at high galactic latitudes.

Its spectral energy distribution (SED) can be empirically treated as a modified blackbody law, described by:

$$I_\nu^{\mathrm{d}}(\vec{n}) = A_{\mathrm{d}}(\vec{n}) \left(\frac{\nu}{\nu_0^{\mathrm{d}}}\right)^{\beta_{\mathrm{d}}} \frac{B_\nu^{\mathrm{Pl}}(T_{\mathrm{d}})}{B_{\nu_0^{\mathrm{d}}}^{\mathrm{Pl}}(T_{\mathrm{d}})}, \tag{11}$$

where $\beta_{\mathrm{d}}(\vec{n})$ represents the dust spectral index, $B_\nu^{\mathrm{Pl}}(T_{\mathrm{d}})$ is the blackbody law at the dust temperature T_{d} and at frequency ν, given by equation (1), and $A_{\mathrm{d}}(\vec{n})$ represents the intensity of the dust signal at the reference frequency ν_0^{d}.

The polarization of dust emission is perpendicular to the magnetic field and can reach up to 25% of the total intensity of dust emission [11].

In general, in studies of CMB polarization and its B-modes, and particularly in the work presented in this document, we will consider dust and synchrotron as the only significant sources of foreground contamination.

[4] In polarization, it is mainly the dust emission from our own Galaxy that constitutes a significant source of contamination for CMB observations.

4 Moments Formalism

4.1 Moment Expansion for an Intensity Signal

Observational studies have shown that the spectral parameters of synchrotron and dust, such as their spectral index and dust temperature, can vary across the sky [16]. Additionally, it is expected that physical conditions within our Galaxy, such as temperature, dust composition, the distribution of relativistic electrons, and other factors, vary in three dimensions.

It is therefore inevitable that these spatial variations in spectral parameters lead to non-trivial spectral mixing. This spectral mixing can occur along the line of sight, between different lines of sight (for example, for an instrumental lobe), or even when calculating spherical harmonic decomposition coefficients. This is because we observe the sum of nonlinear SEDs, which no longer corresponds to the original SED.

For instance, for synchrotron emission, we saw in the previous section that the synchrotron SED follows a power law. However, the spectral index of synchrotron can vary between different lines of sight. This variation in spectral index leads to a distortion in the SED, as the sum of power laws with different spectral indices does not result in a simple power law.

To address these spectral distortions, several main approaches exist for appropriately fitting foreground emission data. Various authors [17, 18] propose dividing the sky into several regions and fitting a power law in each of these regions, thereby minimizing the variation in spectral index between each line of sight. However, this approach has limitations. On one hand, it requires modeling the foregrounds in different regions of the sky where they vary little, which adds a large number of parameters. On the other hand, this method underestimates spectral distortions that will need to be accounted for in future missions.

A second approach involves directly modeling these spectral distortions, as is done with the moment-based approach. Here, we propose to follow [19] and use the moment expansion they propose to model the spectral distortions of the foreground SEDs. Inspired by Taylor expansion, we can write, in general, a second-order expansion as a function of a spectral parameter β (β_s for synchrotron, β_d or T_d for dust) and its derivatives:

$$I_\nu(\beta) = I_\nu(\bar{\beta}) + (\beta - \bar{\beta})\,\frac{\partial I_\nu}{\partial \beta}\Big|_{\beta=\bar{\beta}} + \frac{(\beta-\bar{\beta})^2}{2}\,\frac{\partial^2 I_\nu}{\partial \beta^2}\Big|_{\beta=\bar{\beta}} + 0\left((\beta-\bar{\beta})^3\right), \tag{12}$$

where $\bar{\beta}$ represents the average value of the spectral parameter used as a reference point for the expansion.

We refer to Appendix E for the detailed development of the moment expansion on a pedagogical case of the sum of two power laws with different spectral indices. For example, we find for the moment expansion of the synchrotron spectral index (note that this development also applies to the dust spectral index, see Eq. (11)):

$$I(\nu,\vec{n}) = \frac{I_\nu}{I_{\nu_0}}\left[A(\vec{n}) + \sum_{i=1}^{+\infty}\frac{1}{i!}\omega_i(\vec{n})\ln^i\left(\frac{\nu}{\nu_0}\right)\right], \tag{13}$$

where the coefficients ω_i^n are called moments because they can be identified with the statistical moments of order n of the distribution of $\Delta\beta$, if the amplitudes A_i are interpreted as weights of the probability distribution.

For the temperature expansion, we simply need to differentiate the modified blackbody (MBB) defined by equation (11) with respect to temperature, applying the relation in (12). We will not proceed with the details of this derivation in this study, but they can be found in [20].

4.2 Generalization of Moment Expansion to Polarization

In the previous section, we neglected the intrinsic nature of polarization. In reality, polarization is represented by a complex number:

$$\mathcal{P}_\nu = Q_\nu + iU_\nu = P_\nu e^{2i\psi}, \tag{14}$$

where $\mathcal{P}_\nu$ is the linear polarization spinner, ψ is the polarization angle, and P_ν is the polarized intensity.

This representation describes a spin-2 object, which is appropriate since polarization remains unchanged under a rotation of 180°.

Analogously to the moment expansion in intensity, we can write, to second order, the expansion of the polarization spinner of the spectral index of synchrotron (also applicable to the dust spectral index) proposed by [21]. We refer to Appendix E for the details of the development.

$$\mathcal{P} = \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(\mathcal{W}_0 + \mathcal{W}_1^\beta \ln\left(\frac{\nu}{\nu_0}\right) + \frac{1}{2}\mathcal{W}_2^\beta \ln^2\left(\frac{\nu}{\nu_0}\right) + o\left((\Delta\beta_i)^3\right)\right), \tag{15}$$

where we have defined:

$$\mathcal{W}_0 = \sum_{j=1}^{2} A_j e^{2i\psi_j} \text{ and } \mathcal{W}_n^\beta = \sum_{j=1}^{2} A_j e^{2i\psi_j} \left(\Delta\beta_j\right)^n, \tag{16}$$

where $\mathcal{W}_i$ are the moments analogous to the previous section but this time, complex.

In this case, the polarization angle can be written as:

$$\psi(\nu) \approx \psi(\nu_0) + \frac{\bar{\beta}_\mathrm{I}}{2} \ln\left(\frac{\nu}{\nu_0}\right), \tag{17}$$

where $\psi(\nu_0)$ is the phase of the moment 0 at frequency ν_0. We expect to observe the polarization angle depending on frequency.

In summary, moment expansion is a powerful method because it allows us to describe any mixture of SEDs with a relatively small number of parameters. By capturing the variations of spectral parameters around a mean value, this technique offers increased flexibility and accuracy in analyzing foreground emissions, thus facilitating better separation of components in CMB data.

It has been demonstrated by [21] that moment expansion can be applied to the C_ℓ coefficients and that the BB spectrum corresponds to the imaginary part of this expansion. It also shows that the BB spectrum can be treated independently as an intensity. This approach will be adopted in our study.

Part III

Construction and Validation of the Data Analysis Chain

One of the main limitations in measuring primordial B modes is the precision with which we can separate them from foreground emissions. During this internship, I focused on this crucial issue for the success of future CMB missions. I concentrated on the LiteBIRD experiment and constructed a data analysis chain from the production of sky maps to cosmological analysis on the tensor-to-scalar ratio r, including the estimation of angular power spectra and component separation.

5 Sky Maps

To simulate the sky maps as LiteBIRD will measure them, we need to generate the CMB, the foreground emissions, and the instrumental noise of LiteBIRD. The sky maps are produced using the `HEALPix` library [22].

- **The CMB:** We use the `CAMB` software [23] (Code for Anisotropies in the Microwave Background), which calculates theoretical power spectra of the CMB based on a set of cosmological parameters. I used this software to generate the angular power spectra of the CMB for the ΛCDM model fitting the Planck data [24].

  Using the synfast software, which generates realizations of Gaussian maps I, Q, and U from a given angular power spectrum, I created various realizations of the CMB. Each map simulation is a random realization of CMB fluctuations consistent with the statistical properties defined by the C_ℓ.

- **Foregrounds:** The `PySM` software [25], commonly used in the CMB community, allows the generation of sky maps of different components for various foreground models, created from the component separation products of the Planck and WMAP satellites. We introduce here the models used in this study:

  s0: This is a synchrotron emission model built from the WMAP component separation product at 23 GHz [26]. Maps at other frequencies are generated by extrapolating a simple power law with a constant spectral index $\beta_\mathrm{s} = -3.0$ across the entire sky.

  s1: This model is constructed similarly to s0, but the extrapolation to other frequencies is performed using a power law with different spectral indices in each pixel.

  s7: This model is constructed similarly to s0, but extrapolated using a power law with a curved spectral index (the "running" model described earlier in **??**).

  d0: This is a dust emission model constructed from the Planck component separation product at 353 GHz [27]. Maps at other frequencies are generated by extrapolating a simple MBB law defined by equation (11) with a spectral index $\beta_\mathrm{d} = 1.54$ and a temperature $T_\mathrm{d} = 20$ K.

  d1: This model is constructed similarly to d0, but the extrapolation to other frequencies is performed using a MBB law with different spectral indices and dust temperatures in each pixel.

- **Noise:** The instrumental noise of LiteBIRD is generated by a Gaussian realization of white noise, using the sensitivities of the different frequency channels of LiteBIRD as the standard deviation [7]. To obtain unbiased auto-spectra not affected by noise auto-correlation, it is necessary to simulate "half-frequencies." We create two datasets representing the noise with independent Gaussian realizations. Since these realizations are independent, the correlation between these two noise maps will be zero.

From these different components, we generate 22 sets of simulated maps I, Q, and U for the frequency bands of LiteBIRD for each realization.

6 Masks

Similar to cosmological analyses, we apply a mask to our sky map, primarily to exclude the brightest areas near the Galactic plane and towards the Galactic center, in order to avoid any bias in our estimate of r. The choice of the observed sky fraction (f_{sky}) is critical: a value that is too large leads to significant complexity due to the Galactic center, while a value that is too small reduces the amount of data, thus increasing the standard deviation of our estimator for r. We use a mask constructed from Planck data to maximize the visible portion of the sky ($f_{\text{sky}} = 0.7$) while minimizing the influence of emission from the Galactic center, in order to optimize the accuracy of our analysis.

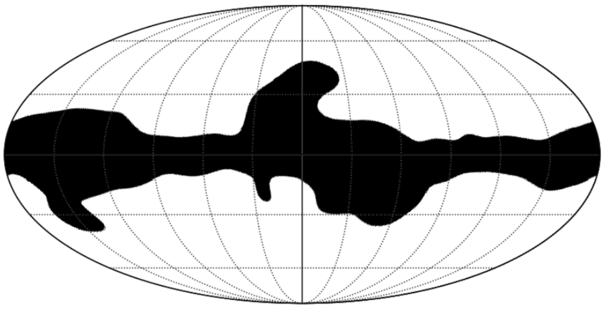


Figure 7: Mask constructed from Planck data with $f_{\text{sky}} = 0.70$. The black area is excluded from our analyses.

7 Power Spectra

7.1 Namaster

Now that we have the sky maps, we calculate their angular power spectra. For this purpose, we use the software `Namaster`[5] [28].

Calculating the power spectrum of a masked map can lead to confusion between E modes and B modes. `Namaster` offers built-in features to account for these mixtures and to unambiguously separate the E and B modes.

I conducted a study on the different parameters proposed by `Namaster` to understand their impact on our results and to optimize our analysis. This test performed on the full sky for the exact power spectrum will also allow us to observe the error generated by the calculation of the angular spectrum as well as the error due to cosmic variance (the uncertainty arising from the fact that we can only observe a single universe and thus a single realization of CMB fluctuations). This result will represent the best precision we can hope to achieve subsequently.

`Namaster` employs a procedure of *apodization* of the masks, which softens their edges by applying a gradual transition rather than a sharp cut-off. `Namaster` offers three types of *apodization*, referred to as *apotypes*:

- **C1**: Uses a cosine-type *apodization* over a finite width. For a distance d from the edge of the mask ω, the *apodized* mask is given by $M(d) = \frac{1}{2}\left(1 + \cos\left(\pi d/\omega\right)\right)$.
- **C2**: Uses the same method as C1 but applies a smoother and more gradual transition.
- **Smooth**: Uses a Gaussian function to apply an extremely smooth and gradual transition.

A second important parameter is the width of the *apodization* transition, named *aposize*. This width influences the smoothness of the transition between masked and unmasked areas. A final parameter is the

[5]https://github.com/LSSTDESC/NaMaster

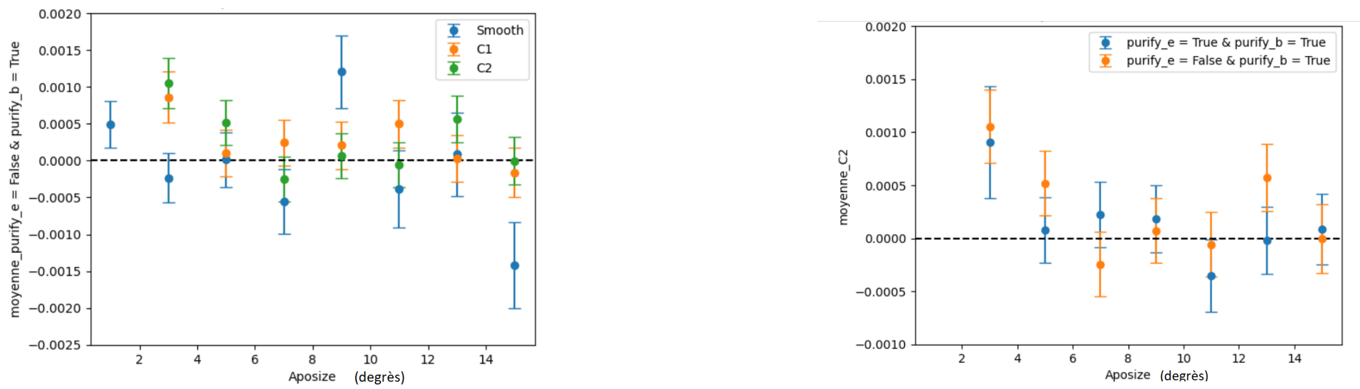
purification of the E and/or B modes. This parameter allows for the separation of the pure E and/or B modes from the *ambiguous* modes, at the cost of increasing their variance.

To conduct this comparative study, we generate CMB maps using the C_ℓ values calculated by the `CAMB` software (see section **??**) assuming $r = 0$, and then we calculate the associated *likelihood* function for this parameter in search of maximum precision while evaluating the bias introduced by the power spectrum calculation. We implement in our model:

$$D_\ell^{\rm CMB} = D_\ell^{\rm lensing} + r \cdot D_\ell^{\rm tensor}, \tag{18}$$

where $D_\ell^{\rm lensing}$ is the angular spectrum of lensing used in our simulations and $D_\ell^{\rm tensor}$ is the spectrum of tensor perturbations for $r = 1$.

We then fit our model to the output data to estimate the parameter r by calculating its *likelihood*. Next, we compare the standard deviations and mean values of this estimator. Throughout the study and subsequent analyses, we binned the ℓ values to make the program more efficient without affecting the results. This analysis allows us to determine the most appropriate parameters for our study.



(a) (b)

Figure 8: Mean and standard deviation of the likelihood posteriors on r for a sky with $r = 0$. The dashed line represents the input value of r at 0. (a) Comparison of the *apotypes* (see legend) as a function of *aposize* for a single purification method (purification of the B modes and not the E modes). (b) Comparison of purification methods (see legend) for *apotype* C2.

Figures 8 illustrate that the Smooth method is less effective. Furthermore, for small *aposize* values, the results obtained with C1 and C2 are biased, while they become equivalent and unbiased for *aposize* values greater than 5 degrees. It is also noted that the choice of purifying the B modes without purifying the E modes, or the choice of purifying both B and E modes, are equivalent, while other choices not shown here are poor.

We find that the error bar is equal to $4 \cdot 10^{-4}$, which results from the combination of cosmic variance from lensing and errors in the calculation of power spectra. This determines a lower limit for the uncertainty that can be reached.

For the rest of our studies, we choose to work with a C2 *apodization* with an *aposize* of 10 degrees and to purify the B modes without purifying the E modes. We make this choice because we are primarily interested in the BB spectra, and it will allow us to compare part of our results with those found in the literature [20].

7.2 Cross-spectra

For each simulation of sky maps, we calculate the angular cross-power spectra $D_\ell^{\rm BB}(\nu_i \times \nu_j)$ from the 22 frequency maps of LiteBIRD. Each of these cross-spectra is associated with a reduced frequency $\nu_{\rm red} = \sqrt{\nu_i \nu_j}$. Using these cross-spectra, we construct the SED from the 253 cross-spectra generated from the different frequency combinations[6] for each multipole bin ℓ.

[6] For $N_{\rm freq}$ frequencies, the number of combinations that can be constructed is $N_{\rm combin} = \frac{N_{\rm freq}(N_{\rm freq}+1)}{2}$

By using the cross-spectra that represent the correlations between different frequencies, we account for the cross-information between the frequency bands, which enhances the sensitivity of our analysis and decorrelates the effects due to spectral mixing.

7.3 Modeling the SED

From these 253 cross-spectra, $D_\ell^{\text{simulation}}$, we aim to model the obtained SED. To do this, we use an algorithm named mpfit [29], which employs the Levenberg-Marquardt method for minimizing χ^2[7]. In this exploratory work, we assume that the two adjacent multipoles are uncorrelated, allowing them to be considered independent. With this approximation, we can minimize χ^2 for each ℓ as follows:

$$\chi^2_{\text{red}}(\ell) = \frac{\left[D_\ell^{\text{simulation}} - D_\ell^{\text{BB}}\right]^T \mathbb{C}^{-1} \left[D_\ell^{\text{simulation}} - D_\ell^{\text{BB}}\right]}{N_{\text{DOF}}}, \tag{19}$$

where N_{DOF} is the number of degrees of freedom, $\mathbb{C}$ is the covariance matrix computed across all realizations, and $D_\ell^{\text{BB}} = D_\ell^{\text{foreground}} + D_\ell^{\text{CMB}}$, our fitting model.

8 Fitting the SED of the Cross-Spectra

To model the SED, we seek to fit parameters consistent with our input simulations, which can be decomposed into CMB and foregrounds. Our study focuses on the polarization of the CMB, although we treat the *BB* spectrum independently. We justify this approach based on prior work presented by [21].

- **CMB Model** The angular power spectra of the CMB are modeled in the same way as for the parameter study in `Namaster`, using the model defined by equation (18).

- **Foreground Model** To fit the SED of the foregrounds, we construct it from the SEDs mentioned in equations (10) for synchrotron emission and (11) for dust emission, but for the angular power spectra. Our foreground emission model then becomes, for each cross-spectrum $D_\ell^{\text{foreground}}(\nu_i \times \nu_j)$:

$$\begin{aligned} D_\ell^{\text{foreground}}(\nu_i \times \nu_j) = {} & D_\ell^{A_\text{d} \times A_\text{d}} \frac{I^\text{d}_{\nu_i}(\beta_\text{d}, T_\text{d}) I^\text{d}_{\nu_j}(\beta_\text{d}, T_\text{d})}{\left(I^\text{d}_{\nu_0}(\beta_\text{d}, T_\text{d})\right)^2} + D_\ell^{A_\text{s} \times A_\text{s}} \frac{I^\text{s}_{\nu_i}(\beta_\text{s}) I^\text{s}_{\nu_j}(\beta_\text{s})}{\left(I^\text{s}_{\nu_0}(\beta_\text{s})\right)^2} \\ & + \frac{D_\ell^{A_\text{d} \times A_\text{s}}}{I^\text{s}_{\nu_0}(\beta_\text{s}) I^\text{d}_{\nu_0}(\beta_\text{d}, T_\text{d})} \left(I^\text{s}_{\nu_j}(\beta_\text{s}) I^\text{d}_{\nu_i}(\beta_\text{d}, T_\text{d}) + I^\text{s}_{\nu_i}(\beta_\text{s}) I^\text{d}_{\nu_j}(\beta_\text{d}, T_\text{d}) \right), \end{aligned} \tag{20}$$

  where $I^\text{d}_{\nu_i}(\beta_\text{d}, T_\text{d})$ and $I^\text{s}_{\nu_i}(\beta_\text{s})$ denote the intensities of dust and synchrotron emission at frequency ν_i. $D_\ell^{A_\text{d} \times A_\text{d}}$, $D_\ell^{A_\text{s} \times A_\text{s}}$ represent the power spectrum of the amplitudes of dust and synchrotron with themselves, and $D_\ell^{A_\text{s} \times A_\text{d}}$ represents the power spectrum of the cross amplitudes between synchrotron and dust. This quantifies the correlation between fluctuations in synchrotron and dust.

- **Models with Moment Expansion** As previously noted, the emissions from dust and synchrotron can exhibit spectral complexity that may introduce a bias in the estimate of r. To counter this issue, we can construct a model based on moment expansion.

  Equation (20) represents the 0th order of this moment expansion; it is possible to incorporate higher-order terms for parameters chosen from β_s, β_d, and T_d, depending on the complexity we wish to model.

[7]The Levenberg-Marquardt minimization is an extension of the least squares method that combines both gradient descent (an iterative method for finding the minimum of a function by following the direction of the descending slope) and the Newton-Raphson method (which relies on the first and second derivatives of the objective function to solve nonlinear optimization problems)

Referring to the SED in equation (13) describing the moment expansion of the spectral index of synchrotron or dust, we can transform it into spherical harmonic space. The model of the cross-spectrum is then given by:

$$\begin{aligned} D_\ell(\nu_1 \times \nu_2) = \frac{I_{\nu_j} I_{\nu_i}}{(I_{\nu_0})^2} \Bigg[D_\ell^{A\times A} + \sum_{i=1}^{+\infty} \frac{1}{i!} D_\ell^{A\times\omega_i} \left(\ln^i\left(\frac{\nu_1}{\nu_0}\right) + \ln^i\left(\frac{\nu_2}{\nu_0}\right)\right) \\ + \sum_{i,j=1}^{+\infty} \frac{1}{i!j!} D_\ell^{\omega_i\omega_j} \left(\ln^i\left(\frac{\nu_1}{\nu_0}\right) \ln^j\left(\frac{\nu_2}{\nu_0}\right)\right)\Bigg]. \end{aligned} \tag{21}$$

This development models all spectral distortions up to order 2, but only contributes to a single β, the spectral index of either synchrotron or dust.

In our case, we must perform the moment expansion of the spectral index of synchrotron and/or the spectral index of dust and/or the temperature of the dust.

When we perform the expansion with multiple parameters, cross terms appear. We refer to Appendix E for the complete formalism we use, where we detail each term of the expansion.

9 Results and Discussion

Our primary goal is to analyze how accurately we can estimate the value of r from sky maps as LiteBIRD will observe them. [20] explored the complexity of dust emission and successfully modeled it using moment expansion. Initially, we will focus on examining the impact of the complexity of synchrotron emission on the parameter r, an analysis that has not been conducted before. Subsequently, in collaboration with Léo Vacher, we have examined a model that integrates the complexity of synchrotron and dust emissions to explore the correlation between these two phenomena.

9.1 Null-Test: s0d0

In this section, we will begin by testing the *null hypothesis* (null-test) where we model the polarized emission of synchrotron and dust using the `PySM` s0 and d0 models, which do not have spectral complexity. The aim of this simulation is to verify whether our fitting parameters are consistent with our initial simulations and to calibrate the minimum precision we can achieve in estimating the parameter r in the presence of foregrounds in this simple case.

To obtain an accurate estimate of r, we proceed as follows: for each value of ℓ, we estimate the free parameters of the model. This process is repeated N_{sim} times, corresponding to the number of simulations. Then, for each ℓ, we generate a histogram from the parameter adjustments and fit a Gaussian to each histogram.

The final result of our estimator for r is obtained by multiplying all the Gaussians associated with each ℓ, followed by fitting a final Gaussian to the resulting distribution, under the assumption that the ℓ bins are statistically *independent*. The mean and standard deviation of this final Gaussian are considered the final result of our estimation of r.

These means and standard deviations are calculated for a fitting of the foreground model without and with moments, respectively, using the models provided in equations (20) and (21), to measure the impact of additional parameters corresponding to the moments on the error associated with the parameter r.

In the case of order 0 (without moment expansion), we have 7 free parameters to estimate for each ℓ: r, β_{s}, β_{d}, T_0, $D_\ell^{A_{\mathrm{s}}\times A_{\mathrm{s}}}$, $D_\ell^{A_{\mathrm{d}}\times A_{\mathrm{d}}}$, $D_\ell^{A_{\mathrm{s}}\times A_{\mathrm{d}}}$.

Fitting at Order 0: We fitted the foreground model and estimated r in an unbiased manner, yielding $r = (0.02 \pm 4.32) \times 10^{-4}$, as illustrated in Figure 9. This result is promising as it is significantly lower than the sensitivity targeted by LiteBIRD ($\sigma(r) \approx 10^{-3}$), thus providing us with the opportunity to include moments for more complex models. We refer to Appendix F for the parameter adjustments of the foregrounds.

Fitting at Order 1 and 2 of the Synchrotron Spectral Index: For each ℓ, we estimated 10 parameters at order 1 and 14 parameters at order 2, in addition to the 7 at order 0. At order 1, we found $r = (0.44 \pm 5.22) \times 10^{-4}$, which does not represent a significant increase in error compared to order 0 while remaining consistent with our previous result. However, at order 2, the standard deviation increases considerably, and our estimator is equal to $(r = 0.16 \pm 2.74) \times 10^{-3}$, which far exceeds the sensitivity targeted by LiteBIRD. This increase may be explained by a strong correlation between certain free parameters of order 2 and the characteristics of the CMB, significantly increasing the standard deviation of the estimator for r. We plan to explore this correlation further using methods such as Markov Chain Monte Carlo to analyze the correlations between parameters.

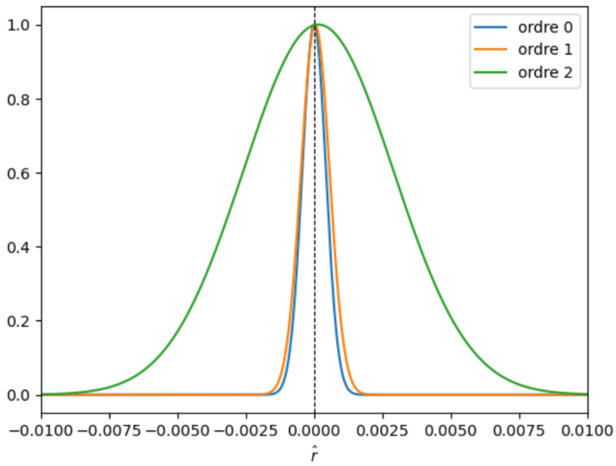


Figure 9: Likelihood of r at orders 0, 1, and 2 for the s0d0 model with $r = 0$

9.2 Complex Synchrotron Models: s1d0 and s7d0

We extended our analysis to more complex synchrotron emission models, namely the s1 and s7 models. Following the same methodology as for s0d0, we examined the impact of this additional complexity on our results for s1d0 and s7d0. The results can be found in Figures 10, and we provide a summary of all our results in Appendix G.

Despite the increased complexity of the models, our analyses did not reveal any significant bias in the obtained estimates. However, it is noteworthy that the s7d0 model at order 0 shows a shift of nearly 1 σ, which disappears when we include order 1. Although our order 1 model is not a *running* model, it effectively captures this feature with a precision similar to that targeted by LiteBIRD.

These observations suggest that the spectral complexity associated with synchrotron emissions combined with simple dust models, such as d0, is relatively low. Therefore, it is possible to forgo the moment expansion of the synchrotron spectral index for simple dust models like d0. However, our study shows that we cannot ignore the correlation between synchrotron and dust, nor can we assume that β_s is independent of ℓ. On the other hand, dust presents significantly higher brightness, making its spectral complexity much more problematic, for which moment expansion is essential [20].

9.3 Complex Dust Model d1s1

Compared to synchrotron radiation, galactic dust is distinguished by its notable brightness across the sky, making it more predominant in LiteBIRD's observation channels. This greater luminosity renders its spectral complexity more problematic, which has a more significant impact on the estimation of r.

To illustrate this complexity and the effect of moment expansion, we examine the d1s1 model, the results of which are presented in Figure 11. The fitting models used are available in Appendix E.

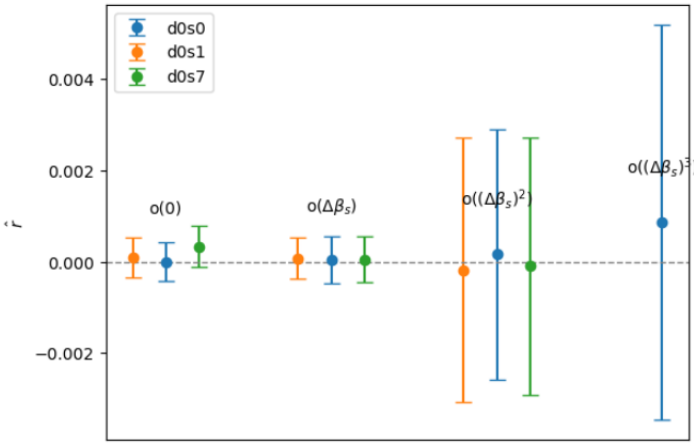


Figure 10: Mean and standard deviation of the likelihood of r for the s0d0, s1d0, and s7d0 models with $r = 0$ at orders 0, 1, and 2.

Without the moment expansion (at order 0), our estimate for the parameter r shows a significant bias of $r = (131 \pm 6) \times 10^{-4}$. However, by including moment expansion for the spectral index and dust temperature, this bias disappears, yielding $r = (0.16 \pm 3.99) \times 10^{-3}$.

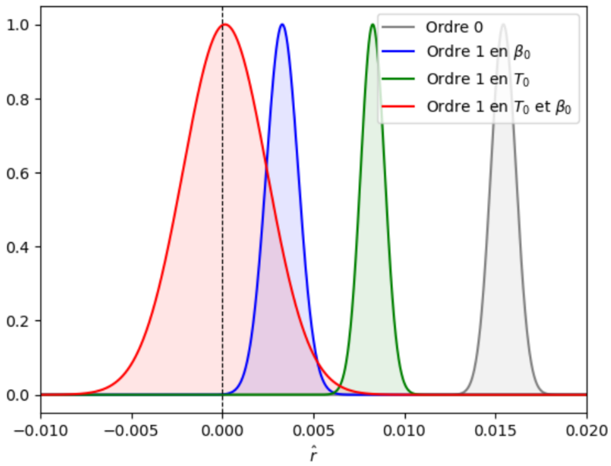


Figure 11: Likelihood of r at order 1 for the s1d1 model with $r = 0$.

It is interesting to note that despite this increased spectral complexity of dust, preliminary studies suggest that the complexity of synchrotron does not play a significant role. However, it is important to emphasize that this conclusion comes from preliminary work, and it would be unwise to draw hasty conclusions.

Nonetheless, it is crucial to note that the introduction of these additional parameters increases the standard deviation of our estimate. At order 1 for β_d and T_d, we obtain $\hat{r} = (1.6 \pm 23.0) \cdot 10^{-4}$.

We observe that the standard deviation associated with this estimate is much larger than that targeted by LiteBIRD (around 10^{-3}). This discrepancy highlights the importance of a more thorough investigation into the relevance of including all free parameters. For instance, a Monte Carlo study could be undertaken to determine which moments are most correlated with the CMB, as well as a study on undetected moments to exclude them from the analysis.

Complete details on the results of our studies are provided in Appendix G.

Part IV

Conclusion

The observation of foregrounds in our Galaxy reveals SEDs that vary in three dimensions. When observing these signals, several different SEDs are inevitably averaged, which distorts and complicates the integrated SED. The moment expansion formalism of the SED provides a framework to describe this complexity with few parameters, which could be useful for component separation to measure primordial B modes and the tensor-to-scalar ratio r. It has been shown to be an effective modeling approach for dust in the case of LiteBIRD [20]. During the internship whose results are presented in this document, we aimed to explore the interest of this approach for the synchrotron emission of our Galaxy.

To conduct this study, I developed a data analysis pipeline. I constructed sky maps containing different components: CMB with $r = 0$ (containing only the lensing component), foreground components with various complexities, and LiteBIRD instrumental noise. I simulated these maps for the 22 frequency bands of LiteBIRD and calculated all the cross-spectra between these bands. Then, I fitted a model of CMB and foregrounds with and without moment expansion to study the precision with which we could estimate r.

First, I wanted to evaluate the error arising from the method used to calculate the angular power spectra, `Namaster`, from simulations containing only CMB with $r = 0$. I found unbiased results (that is, consistent with 0) and a standard deviation on r comparable to that of cosmic variance, demonstrating that the method does not introduce additional error. Next, I added noise at the expected level for LiteBIRD as well as foregrounds (synchrotron and dust) without spectral complexity, and found that at orders 0 and 1, the estimator of r was unbiased and its error bar was about 5×10^{-4}, which is below the precision targeted by LiteBIRD. However, we noticed a significant increase in the error bar when performing a moment expansion at order 2, yielding $r = (0.16 \pm 2.74) \times 10^{-3}$. This increase causes the error bar to exceed that aimed for by LiteBIRD.

I then explored different synchrotron models with increasing spectral complexity, and the analyses revealed no significant bias in the estimates of r. Furthermore, when studying a model where the spectral index of synchrotron varies with frequency, the likelihood of r is shifted by 1 σ from the input value $r = 0$. Adding the expansion at order 1 in β_s allows for the posterior to be recentered without increasing its standard deviation. Finally, I examined how synchrotron couples with the complexity of dust, and as shown by [20], the complexity of dust has a strong impact on the estimation of r. However, by using moment expansion on the dust MBB parameters, we found that this model adequately captures the complexity of the simulation. In contrast, the complexity of synchrotron does not seem to significantly impact the analysis, which is an important result for component separation.

However, many avenues have been highlighted. Firstly, my programs take a long time to execute (for example, three days for the moment expansion at order 2 in the spectral index of synchrotron). We might consider changing methods, such as using Markov Chain Monte Carlo fitting. This method would also allow for the study of the different correlations between parameters, thus enabling an investigation into the importance of certain correlations and their contributions to the estimation of r. I started studying the impact that moment expansion may have on real data from the QUIJOTE telescope by calculating their angular power spectra. It was observed that the noise was correlated between certain frequency bands. I would like to deepen this analysis at the end of my internship, as it would help characterize the spectral complexity of the actual sky in the data, to see if the synchrotron models we used in our analysis are realistic or not.

The precise understanding and separation of the CMB components will play a central role in future cosmological research. The detection of B modes would herald a revolution in our understanding of the primordial Universe.

Appendix

A The Λ-CDM Model

In this section, we discuss the foundations of the Λ-CDM model. This model serves as the cornerstone of our current understanding of the observable Universe, and its predictions provide the theoretical framework necessary to interpret observational data, particularly for the CMB and its polarization.

A.1 The FLRW Metric

The standard cosmological model that best describes the evolution of the Universe on large scales is the Λ-CDM model. This model is based on key concepts of modern cosmology, such as the cosmological constant (Λ) and cold dark matter (*"Cold Dark Matter"*).

To understand in detail the dynamics of the Universe and its observable implications, we adopt the Friedmann-Lemaître-Robertson-Walker ("FLRW") metric, which we assume for this study[8], described by:

$$ds^2 = -dt^2 + a(t)^2\left(\frac{dr^2}{1-kr^2} + r^2\left(d\theta^2 + \sin^2(\theta)d\phi^2\right)\right) \tag{22}$$

In this equation, $a(t)$ represents the scale factor describing the distances between objects in the Universe, and k represents the spatial curvature, which can take the values 0, 1, or -1 for a flat, spherical, or hyperbolic Universe, respectively.

A.2 The Friedmann Equations

By deriving Einstein's equations and using the FLRW metric, we obtain the Friedmann-Lemaître equations, represented by:

$$\begin{aligned} H^2 := \left(\frac{\dot{a}}{a}\right)^2 &= \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \\ \frac{\ddot{a}}{a} &= \frac{-4\pi G}{3}(\rho + 3P) + \frac{\Lambda}{3} \end{aligned} \tag{23}$$

In these equations, ρ represents the total energy density of the Universe, P is the pressure, and H is the Hubble parameter. Λ is the cosmological constant and k represents the spatial curvature.

These equations describe the evolution of the Universe based on its matter and energy density.

We introduce the parameters $\rho_k = -3k/(8\pi G a^2)$ and $\rho_\Lambda = 3H^2/(8\pi G)$ for the curvature density and the density of the cosmological constant, respectively. Additionally, $\Omega_i = \rho_i/\rho_c$ is defined as the cosmological parameter, where $\rho_c = 3H^2/(8\pi G)$ is the critical density of a flat universe without a cosmological constant.

We rewrite the first Friedmann equation by decomposing $\rho = \rho_r + \rho_m$, representing the energy density of radiation and matter, as:

$$\Omega_\mathrm{r} + \Omega_\mathrm{m} + \Omega_\Lambda + \Omega_\mathrm{k} = 1 \tag{24}$$

In addition to these equations, we write a continuity equation for a primordial fluid, whose derivation is omitted in this report[9].

[8] The FLRW metric is based on the assumption that the Universe is homogeneous and isotropic on large scales, an assumption largely confirmed by cosmological observations such as the large-scale distribution of galaxies.

[9] This equation can be derived from the two Friedmann equations or from the conservation of the energy-momentum tensor.

$$\dot{\rho} + 3H(\rho + 3P) = 0 \tag{25}$$

We also define an equation of state for this primordial fluid.

$$\omega = \frac{P}{\rho} \tag{26}$$

Using the equation of state, we obtain a solution to the continuity equation (25), $\rho \propto a^{-3(1+\omega)}$. These equations (26) and (25) allow us to study the equation of state for the different contents of the Universe.

- **Matter:** The condition $E_{\text{kinetic}} \ll E_{\text{mass}}$ characterizes matter, implying $P = 0 \Rightarrow \omega = 0$.
- **Radiation:** In contrast, radiation is defined by the condition $E_{\text{kinetic}} \gg E_{\text{mass}}$, leading to the null trace condition of the energy-momentum tensor, generally defined by:

$$T^{\mu\nu} = (\rho + P)u^{\mu}u^{\nu} - pg^{\mu\nu} \tag{27}$$

  Thus, $\mathrm{Tr}\,(T^{\mu\nu}) = 0 \Rightarrow \rho = 3P \Rightarrow \omega = \frac{1}{3}$

- **Cosmological Constant:** An equation of state for the cosmological constant can be established from the energy-momentum tensor given by:

$$T^{\mu\nu} = \begin{pmatrix} \frac{-\Lambda}{8\pi G} & 0 & 0 & 0 \\ 0 & \frac{-\Lambda}{8\pi G} & 0 & 0 \\ 0 & 0 & \frac{-\Lambda}{8\pi G} & 0 \\ 0 & 0 & 0 & \frac{-\Lambda}{8\pi G} \end{pmatrix} \tag{28}$$

  Using the definition of this tensor given by equation (27), we deduce that $P = -\rho \Rightarrow \omega = -1$.

The model we have just presented provides the essential theoretical framework for understanding the origin and evolution of the Universe. In the context of our study of the B modes of the CMB, it is crucial because it predicts the fundamental characteristics of temperature and polarization anisotropies due to primordial perturbations presented in Appendix C. In particular, the detection of primordial B modes could provide direct evidence for inflation, which we will study in more detail in Appendix B.

B Inflation

In this section, we introduce the motivations and a simple model of inflation. This model leads to predictions regarding the B modes, thus paving the way for their measurement and analysis.

B.1 Motivations for Inflation: Limits of the Hot Big Bang Model

The Big Bang hypothesis has transformed our understanding of the Universe by proposing a coherent model for its origin. Despite its remarkable predictive power, the increasing precision of cosmological observations has highlighted new problems. Here, we will examine two of these limitations[10]

- **Flatness Problem:** Observations of the CMB suggest that our Universe has a geometry close to flatness (e.g., [30]). Using the Friedmann equations, we can explore the implications at the time of the Planck era. Consider the first Friedmann equation evaluated at the current time $t = t_0$ (noting that the Friedmann equations are valid at all times).

$$\Omega_0 := \Omega_\mathrm{m} + \Omega_\mathrm{r} + \Omega_\Lambda = 1 + \frac{k}{a_0^2 H_0^2} \tag{29}$$

  Furthermore, the hot Big Bang model suggests that the scale factor evolves over time according to a power-law $a(t) \propto t^p$, with p 1.

  If we examine the second term of equation (29), we note that $\left(a^2\left(\dot{a}/a\right)^2\right)^{-1} \propto t^{2(1-p)} > 0$, Therefore, $\Omega_0 - 1$ grows with time.

  When we go back to the Planck time: $t_\mathrm{p} \approx 10^{-43}$ s, we obtain $\Omega_0(t_\mathrm{p}) \approx 1 + 10^{-60}$, which implies that without a very fine adjustment at the time of the Big Bang, the flatness observed today would be very unlikely, raising the flatness problem as a "fine-tuning" problem.

- **Horizon Problem:** Due to the finite speed of light, we can define the particle horizon as the largest distance from which a light ray can reach an observer during the age of the Universe. Consider the trajectory of a photon on the light cone disregarding its angular dependence, defined by $a\,dr = dt$. We then define a comoving radius (accounting for the expansion of the Universe) as follows:

$$R_\mathrm{comoving} = \int_{t_\mathrm{i}}^{t} \frac{dt'}{a(t')} \tag{30}$$

  The physical quantity associated with this comoving radius is given by $d_H(t) = a(t)R_\mathrm{comoving}$. Thus, using the temporal dependence of the scale factor predicted by the hot Big Bang model in the form of a power law, we find a dependence of d_H of the form $t/(1-p)$ (taking $t_i = 0$, the initial time at the moment of the Big Bang). This implies that the comoving radius grows with time as $t^{1-p}/(1-p)$.[11] Thus, the fraction of the Universe in causal contact increases over time.

  CMB observations raise a question: How is it that regions of the Universe that have never been in direct causal contact exhibit homogeneous characteristics such as temperature?

These enigmas have motivated the search for solutions, such as cosmic inflation, which will be discussed in the next section, proposing an alternative and coherent perspective on the early moments of the Universe.

[10] The formation of structures and the magnetic monopole problem are also additional issues.

[11] Meanwhile, note that the temporal dependence of $(aH)^{-1} \propto t^{1-p}/p \approx R_\mathrm{comoving}$. We will take the quantity $(aH)^{-1}$ as a reference for the comoving radius.

B.2 The SFSR ("Single-Field Slow-Roll") Inflationary Model

The introduction of inflation within the framework of the Big Bang model provides an elegant solution to these problems. This theory emerged in the 1980s, initially by Alan Guth and Andrei Linde, and was developed by postulating a brief period of exponential expansion in the early moments of the Universe [31, 32]. It was introduced in such a way that the Hubble radius decreases during the primordial phase of the Universe, implying a period where $R_{\text{comoving}} \approx (aH)^{-1} = \dot{a}^{-1}$ decreases. This means that $\ddot{a}\ 0$, indicating primordial acceleration of the Universe.

By examining the continuity equation, the condition on $\ddot{a}$ translates to:

$$\rho + 3P < 0 \Rightarrow \omega < \frac{-1}{3} \tag{31}$$

A scalar field ϕ called the *inflaton* can be described by this exotic equation of state. By using the FLRW metric and starting from the expressions for energy density $\rho = (1/2)\dot{\phi}^2 + V(\phi)$ and pressure $P = (1/2)\dot{\phi}^2 - V(\phi)$[12], we find the dynamic equation of inflation:

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0, \tag{32}$$

where $\dot{\phi} = \partial\phi/\partial t$ and $V' = \partial V/\partial\phi$.

For the condition (31) to be satisfied, it is necessary that $(1/2)\dot{\phi}^2\ 2V(\phi)$.

To deepen our analysis, we can revisit the Friedmann equations while considering the slow-roll regime, where the kinetic term in the first Friedmann equation is negligible compared to the potential, and where the acceleration term is negligible compared to the derivative of the potential in the second equation. Thus, we obtain the equations:

$$H^2 = \frac{8\pi G}{3}\left(\frac{1}{2}\dot{\phi}^2 + V\right) \approx \frac{8\pi G}{3}V \tag{33}$$

$$\ddot{\phi} + 3H\dot{\phi} + V' \approx 3H\dot{\phi} + V = 0$$

Using the two conditions of the slow-roll regime, we can define two parameters that must satisfy the conditions stated in (34) based on the Friedmann equations given by:

$$\frac{1}{2}\dot{\phi}^2 \ll V \Rightarrow \epsilon := \frac{M_{\text{pl}}^2}{2}\left(\frac{V'}{V}\right)^2 \ll 1 \tag{34}$$

$$\ddot{\phi} \ll V' \Rightarrow \eta := \frac{-\ddot{\phi}}{H\dot{\phi}} \ll 1$$

where $M_{\text{pl}} \approx 1/(\sqrt{8\pi G})$ represents the Planck mass. We can show that $\eta \approx M_{\text{pl}}^2 V''/V$. Thus, we reduce to a problem depending solely on the potential energy V. The observables from future cosmological observation missions will need to provide data to recover the value of the potential V (see Appendix C.2).

[12] By deriving the action of a scalar field given by $S_\phi = \int d^4x\sqrt{-g}\left(-\frac{1}{2}\partial^\mu\phi\partial_\mu\phi - V(\phi)\right)$, where $g = \det(g^{\mu\nu})$

C Perturbations

C.1 Density Perturbations

In this section, in a non-exhaustive manner, we briefly introduce cosmological perturbations in order to introduce the quantity r, corresponding to the ratio of scalar to tensor perturbations, whose imprint can be observed in the CMB.

If the scalar field ϕ and the expansion rate H can vary in time, then they can also vary in space. The fluctuations in ϕ lead to fluctuations in the energy density ρ, which in turn induce fluctuations in the metric.

The most general perturbed metric has 10 components. Among these, 4 are non-physical gauge degrees of freedom, 2 are tensor degrees of freedom, 2 are vector degrees of freedom, and the last 2 can be reduced to a single scalar perturbation.

In the comoving gauge, the components of the metric can be written as follows:

$$\begin{gathered} g_{0\mu} = 0 \\ g_{ij} = a^2(t)\exp\left(2\mathcal{R}(\vec{x},t)\right)\delta_{ij}, \end{gathered} \tag{35}$$

where $\mathcal{R}(\vec{x},t)$ represents the curvature perturbations.

We can apply these conditions to the Einstein-Hilbert action[13] S_{EH} combined with the action of the scalar field $S_{\mathcal{R}}$, defined respectively by the equations (36) and (37).

$$S_{\mathrm{EH}} = \frac{1}{16\pi G}\int d^4x\sqrt{-g}R, \tag{36}$$

where g represents the determinant of the metric and R the Ricci tensor (defined as the contraction of the Riemann tensor, representing the curvature of spacetime).

$$S_{\mathcal{R}} = \int dt \int d^3x\, a^3(t)\left[\frac{1}{2}\dot{v}^2 - \frac{1}{2a^2}\left(\nabla v\right)^2\right] \tag{37}$$

where $v^2 = 2M_{\mathrm{Pl}}^2\epsilon\mathcal{R}$, a function of ϵ defined in (34) and the curvature perturbations $\mathcal{R}$. For more information, see [5].

To study the fluctuation modes of the field at different frequencies, we move to momentum space by performing a Fourier transform of the action of the scalar field, yielding:

$$S_{\mathrm{R}} = \sum_k \int dt\, a^3\left[\frac{1}{2}|\dot{v}_k|^2 - \frac{1}{2}\left(\frac{k}{a}\right)^2|v_k|^2\right] \tag{38}$$

We recognize in the action given in (38) a sum of actions of a set of decoupled oscillators of Fourier modes k, which admit the following solution:

$$\ddot{v}_k + 3H\dot{v}_k + \left(\frac{k}{a}\right)^2 v_k = 0 \tag{39}$$

$$\Rightarrow v_k(t) = \frac{H}{(2k^3)^{1/2}}\left(i + \frac{k}{aH}\right)e^{-\frac{ik}{aH}} \Rightarrow |v_k|^2 \xrightarrow{k\ll aH} \frac{H^2}{2k^3}$$

After crossing the horizon ($k \ll aH$), inflationary expansion converts quantum fluctuations of curvature into classical super-horizon curvature perturbations (distances such that their influence cannot be

[13]This action can be derived from the principle of least action, showing that the Einstein field equations, describing the dynamics of spacetime in response to the presence of mass and energy, follow from it.

felt or observed in our observable universe). These perturbations subsequently manifest in the CMB as anisotropies, shaping the structure of the future Universe.

Moreover, it is important to note that the primordial perturbations are a realization of a random field for each Fourier amplitude. We can express this amplitude as $\mathcal{R}_k^2 = v^2/(2M_{\text{pl}}^2\epsilon)$, where a Gaussian distribution of variance is selected, as established in:

$$\langle|\mathcal{R}_k|^2\rangle = \frac{H^2}{4M_{\text{pl}}^2\epsilon k^3} \tag{40}$$

We can then define the curvature power spectrum, denoted $\Delta^2_{\mathcal{R}}(k)$[14], as indicated in the equation:

$$\Delta^2_{\mathcal{R}}(k) := \frac{k^3}{2\pi^2}\langle|\mathcal{R}_k|^2\rangle = \frac{1}{8\pi^2}\frac{H^2}{M_{\text{Pl}}^2\epsilon} \approx \frac{1}{24\pi^2}\frac{V}{M_{\text{Pl}}^4\epsilon} \tag{41}$$

With V the energy scale of inflation using the slow-roll regime approximation in (33).

This definition allows us to study the distribution of density perturbations at different length scales, which is essential for determining r, the ratio between tensor and scalar perturbations.

C.2 Primordial Gravitational Wave Perturbations

Similarly to the solutions of Maxwell's equations in the absence of a source, which describe the propagation of photons with two linear polarizations, a solution of Einstein's equations describes the propagation of gravitational waves with two polarizations denoted $+$ and $\times$.

We place ourselves in a perturbed FLRW metric defined by:

$$g_{ij} = a^2(t)(\delta_{ij} + 2h_{ij}) \tag{42}$$

where h_{ij} represents the perturbation of the metric. We can use the *TT* ("Transverse Traceless") gauge defined by the conditions $\partial^i h_{ij} = h^i_i = 0$.

In a similar way to scalar perturbations, we can write the action of a tensor field and move to Fourier space, as indicated in the equation:

$$\begin{aligned} S_{\text{h}} &= \frac{1}{4}\int dt \int d^3x a^3 M_{\text{pl}}\left[\frac{1}{2}(\dot{h}_{ij})^2 - \frac{1}{2a^2}(\partial_k h_{ij})^2\right] \\ \xrightarrow{\text{TF}} S_{\mathcal{R}} &= \sum_{p=+,\times}\sum_k \int dt a^3 \left[\frac{1}{2}|\dot{v}_{p,k}|^2 - \frac{1}{2}\left(\frac{k}{a}\right)^2 |v_{p,k}|^2\right] \end{aligned} \tag{43}$$

with $v_{p,k} = h_p M_{\text{Pl}}/2$, analogous to the field v defined in the previous section. For more information, see [5].

We can then define the tensor power spectrum by:

$$\Delta^2_h(k) := 2\frac{k^3}{2\pi^2}\langle|h_{p,k}|^2\rangle = \frac{2}{\pi^2}\frac{H^2}{M_{\text{Pl}}^2} \tag{44}$$

Finally, using equations (41) and (44), along with the fact that during inflation $H^2 \propto V$, we define the scalar-to-tensor ratio as:

$$r := \frac{\Delta^2_h}{\Delta^2_{\mathcal{R}}} \approx 0.1\left(\frac{V}{[2\times 10^{16}\ \text{GeV}]^4}\right) \tag{45}$$

[14] The spectral index of the Λ-CDM model is defined by $n_s(k) - 1 = d\ln(\Delta^2_{\mathcal{R}}(k))/d\ln(k)$

Thus, by measuring r, it will be possible to infer the energy of the Universe during inflation and thus constrain many inflationary models. The parameter r quantifies the relative amplitude of B modes compared to density modes in the CMB.

D Foreground Emissions

In this Appendix, we will cite the main sources of foreground emissions that we may encounter in the study of CMB maps.

- **Free-free emission:** Due to bremsstrahlung radiation. This emission is generated by the deceleration of electrons scattering off the positively charged ions in the hot interstellar medium (ISM). This emission is very weakly polarized (1%) [33].
- **Anomalous Microwave Emission (AME):** The origin of AME is still poorly understood. A plausible hypothesis suggests that small, rapidly rotating dust grains could emit this radiation [34]. AME is also very weakly polarized (1%).
- **Molecular emission lines:** In the ISM, emission lines from certain molecules (the most significant being CO) fall within the frequency range of CMB observations. Although this emission can be polarized, it is negligible in the study of B modes. For a review, see [35].
- **Cosmic Infrared Background (CIB):** The CIB is a diffuse emission due to many unresolved extragalactic sources. This emission is very weakly polarized (1%) [36].
- **Point sources:** These sources, on the order of an instrumental beam size, can originate from quasars, radio galaxies, pulsars, or the infrared emission from distant galaxies.
- **Zodiacal light:** The zodiacal light is due to the reflection and re-emission of solar radiation by dust grains in our solar system. This emission does not interfere with the analysis of the B modes of the CMB. For a review, see [37].

E Moment Formalism

In this section, we will detail the models we use in this study, along with the details of the moment expansion calculations.

E.1 Pedagogical Case of the Sum of Two Power Laws

To illustrate the details of a moment expansion calculation, let us consider the example of two power laws that we develop up to first order. We consider the sum of two contributions, for example, from two regions along the line of sight with different spectral indices:

$$I_\nu = A_1 \left(\frac{\nu}{\nu_0}\right)^{\beta_1} + A_2 \left(\frac{\nu}{\nu_0}\right)^{\beta_2} \tag{46}$$

By expanding each term around the mean spectral index $\bar{\beta} = (\beta_1 + \beta_2)/2$ and noting $\Delta\beta_i = \beta_i - \bar{\beta}$, we obtain:

$$I_\nu = A_1 \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(1 + \Delta\beta_1 \ln\left(\frac{\nu}{\nu_0}\right)\right) + O\left((\Delta\beta_1)^2\right) + A_2 \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(1 + \Delta\beta_2 \ln\left(\frac{\nu}{\nu_0}\right)\right) + O\left((\Delta\beta_2)^2\right) \tag{47}$$

Finally, by noting $\bar{A} = A_1 + A_2$ and $\omega_1^\beta = A_1/\bar{A} \cdot \Delta\beta_1 + A_2/\bar{A} \cdot \Delta\beta_2$, we obtain:

$$I_\nu = \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left([A_1 + A_2] + [A_1\Delta\beta_1 + A_2\Delta\beta_2] \ln\left(\frac{\nu}{\nu_0}\right) + O\left((\Delta\beta)^2\right)\right) = \bar{A} \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(1 + \omega_1^\beta \ln\left(\frac{\nu}{\nu_0}\right) + O(\Delta\beta_2)^2\right) \tag{48}$$

The coefficients ω_i^n are called moments because they can be identified with the statistical moments of order n of the distribution of $\Delta\beta$, if the amplitudes A_i are interpreted as normalizations of the probability distribution.

As statistical moments, it is possible to interpret ω_1^β as a correction to the pivot mean $\bar{\beta}$. The corrected value $\bar{\beta}_{\text{corr}}$ will be the choice of the pivot spectral index corresponding to the cancellation of the first-order term, as follows:

$$\omega_1^\beta = 0 \Rightarrow A_1 (\beta_1 - \bar{\beta}_{\text{corr}}) + A_2 (\beta_2 - \bar{\beta}_{\text{corr}}) = 0 \Rightarrow \bar{\beta}_{\text{corr}} = \frac{A_1\beta_1 + A_2\beta_2}{\bar{A}} \tag{49}$$

E.2 Generalization of the Moment Expansion to Polarization

In the same way as in the case of the sum of two power laws, we can write:

$$\mathcal{P}_\nu = \mathcal{P}_\nu^1 + \mathcal{P}_\nu^2 = A_1 \left(\frac{\nu}{\nu_0}\right)^{\beta_1} e^{2i\psi_1} + A_2 \left(\frac{\nu}{\nu_0}\right)^{\beta_2} e^{2i\psi_2} \neq \bar{A} \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} e^{2i\bar{\psi}} \tag{50}$$

A new consequence regarding the intensity is that the total phase is not a simple sum of the two initial phases.

We can expand to first order:

$$\begin{aligned} \mathcal{P}_\nu &= A_1 \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} e^{2i\psi_1} \left(1 + \Delta\beta_1 \ln\left(\frac{\nu}{\nu_0}\right) + o\left((\Delta\beta_1)^2\right)\right) + A_2 \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} e^{2i\psi_2} \left(1 + \Delta\beta_2 \ln\left(\frac{\nu}{\nu_0}\right) + o\left((\Delta\beta_2)^2\right)\right) \\ &= \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(\left(A_1 e^{2i\psi_1} + A_2 e^{2i\psi_2}\right) + \left(A_1 e^{2i\psi_1}\Delta\beta_1 + A_2 e^{2i\psi_2}\Delta\beta_2\right) \ln\left(\frac{\nu}{\nu_0}\right) + o\left((\Delta\beta_i)^2\right)\right) \end{aligned} \tag{51}$$

We then obtain, similarly to what we did previously, a generalization proposed by [21]:

$$\mathcal{P}_\nu = \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta}} \left(\mathcal{W}_0 + \mathcal{W}_1^\beta \ln\left(\frac{\nu}{\nu_0}\right) + \frac{1}{2}\mathcal{W}_2^\beta \ln^2\left(\frac{\nu}{\nu_0}\right) + o\left((\Delta\beta_i)^3\right)\right) \tag{52}$$

Where we have defined:

$$\mathcal{W}_0 = \sum_{j=1}^{2} A_j e^{2i\psi_j} \text{ and } \mathcal{W}_n^\beta = \sum_{j=1}^{2} A_j e^{2i\psi_j} \left(\Delta\beta_j\right)^n \tag{53}$$

By requiring the cancellation of the first-order term as we did for the intensity in 49, we notice that the corrected value of the pivot spectral parameter is a complex number:

$$\mathcal{W}_1^\beta = 0 \Rightarrow \bar{\beta}_{\text{corr}} = \frac{A_1 e^{2i\psi_1}\beta_1 + A_2 e^{2i\psi_2}\beta_2}{A_1 e^{2i\psi_1} + A_2 e^{2i\psi_2}} \tag{54}$$

And, by decomposing $\bar{\beta}_{\text{corr}}$ into an imaginary part $\bar{\beta_{\text{I}}}$ and a real part $\bar{\beta_{\text{R}}}$, the correction to the power law of the SED becomes:

$$\left(\frac{\nu}{\nu_0}\right)^{\bar{\beta_{\text{corr}}}} = \left(\frac{\nu}{\nu_0}\right)^{\bar{\beta_{\text{R}}}} e^{i\bar{\beta_{\text{I}}}\ln\left(\frac{\nu}{\nu_0}\right)} \tag{55}$$

Thus, we can interpret the real part of $\bar{\beta}_{\text{corr}}$ as a real correction of the spectral index, and its imaginary part as a first-order correction in terms of the frequency of the polarization angle, which now allows us to predict that:

$$\psi(\nu) \approx \psi(\nu_0) + \frac{\bar{\beta_{\text{I}}}}{2} \ln\left(\frac{\nu}{\nu_0}\right) \tag{56}$$

Where $\psi(\nu_0)$ is the phase of moment 0. Thus, the corrections of the complex spectral parameter naturally fit within the framework of polarization. And we see that the total phase becomes dependent on frequency.

E.3 Synchrotron Moments

Up to second order in the expansion of the spectral index of synchrotron radiation, our model becomes:

$$
\begin{aligned}
D_\ell(\nu_i \times \nu_j) = D_\ell^{A_\mathrm{d}\times A_\mathrm{d}} & \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \\
& + D_\ell^{A_\mathrm{s}\times A_\mathrm{s}} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \\
& + \frac{D_\ell^{A_\mathrm{d}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \right) \\
& + D_\ell^{A_\mathrm{s}\times \omega_1} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) + \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{\omega_1\times \omega_1} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \ln\left(\frac{\nu_i}{\nu_0}\right) \ln\left(\frac{\nu_j}{\nu_0}\right) \\
& + \frac{D_\ell^{A_\mathrm{d}\times \omega_1}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_j}{\nu_0}\right) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_i}{\nu_0}\right) \right) \\
& + D_\ell^{A_\mathrm{s}\times \omega_2} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \left(\ln^2\left(\frac{\nu_i}{\nu_0}\right) + \ln^2\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{\omega_1\times \omega_2} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \left(\ln^2\left(\frac{\nu_j}{\nu_0}\right) \ln\left(\frac{\nu_i}{\nu_0}\right) + \ln^2\left(\frac{\nu_i}{\nu_0}\right) \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{\omega_1\times \omega_2} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \ln^2\left(\frac{\nu_j}{\nu_0}\right) \ln^2\left(\frac{\nu_i}{\nu_0}\right) \\
& + \frac{D_\ell^{A_\mathrm{d}\times \omega_1}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln^2\left(\frac{\nu_i}{\nu_0}\right) + I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln^2\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + \cdots\cdots
\end{aligned}
\tag{57}
$$

E.4 Dust Moments

Up to first order in the expansion of the spectral index of dust and the temperature of dust, our model becomes:

$$
\begin{aligned}
D_\ell(\nu_i \times \nu_j) = D_\ell^{A_\mathrm{d}\times A_\mathrm{d}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \\
+ D_\ell^{A_\mathrm{s}\times A_\mathrm{s}} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \\
+ \frac{D_\ell^{A_\mathrm{d}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \right) \\
+ D_\ell^{A_\mathrm{d}\times \omega_1^{\beta_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) + \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
+ D_\ell^{\omega_1^{\beta_\mathrm{d}}\times \omega_1^{\beta_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
+ D_\ell^{A_\mathrm{d}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\Theta_i + \Theta_j - 2\Theta_0 \right) \\
+ D_\ell^{\omega_1^{T_\mathrm{d}}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left((\Theta_j - \Theta_0)(\Theta_i - \Theta_0) \right) \\
+ D_\ell^{\omega_1^{\beta_\mathrm{d}}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left((\Theta_i - \Theta_0) \ln\left(\frac{\nu_j}{\nu_0}\right) + (\Theta_j - \Theta_0) \ln\left(\frac{\nu_i}{\nu_0}\right) \right) \\
+ \frac{D_\ell^{\omega_1^{\beta_\mathrm{d}}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_i}{\nu_0}\right) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
+ \frac{D_\ell^{\omega_1^{T_\mathrm{d}}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) (\Theta_i - \Theta_0) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) (\Theta_j - \Theta_0) \right) \\
+ \cdots
\end{aligned} \tag{58}
$$

E.5 Moments of Synchrotron and Dust

Up to first order of the expansion in moment of the spectral index of dust and synchrotron as well as that of the dust temperature, by setting:

$$\Theta_\nu = \frac{x}{T}\frac{e^x}{e^x - 1}, \text{ with } x = \frac{h\nu}{kT} \tag{59}$$

Our model becomes:

$$
\begin{aligned}
D_\ell(\nu_i \times \nu_j) = {} & D_\ell^{A_\mathrm{d}\times A_\mathrm{d}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \\
& + D_\ell^{A_\mathrm{s}\times A_\mathrm{s}} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \\
& + \frac{D_\ell^{A_\mathrm{d}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \right) \\
& + D_\ell^{A_\mathrm{d}\times \omega_1^{\beta_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) + \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{\omega_1^{\beta_\mathrm{d}}\times \omega_1^{\beta_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{A_\mathrm{d}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left(\Theta_i + \Theta_j - 2\Theta_0 \right) \\
& + D_\ell^{\omega_1^{T_\mathrm{d}}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left((\Theta_j - \Theta_0)(\Theta_i - \Theta_0) \right) \\
& + D_\ell^{\omega_1^{\beta_\mathrm{d}}\times \omega_1^{T_\mathrm{d}}} \frac{I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d})}{\left(I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})\right)^2} \left((\Theta_i - \Theta_0)\ln\left(\frac{\nu_j}{\nu_0}\right) + (\Theta_j - \Theta_0)\ln\left(\frac{\nu_i}{\nu_0}\right) \right) \\
& + D_\ell^{A_\mathrm{s}\times \omega_1^{\beta_\mathrm{s}}} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \left(\ln\left(\frac{\nu_i}{\nu_0}\right) + \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + D_\ell^{\omega_1^{\beta_\mathrm{s}}\times \omega_1^{\beta_\mathrm{s}}} \frac{I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s})}{\left(I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s})\right)^2} \ln\left(\frac{\nu_i}{\nu_0}\right) \ln\left(\frac{\nu_j}{\nu_0}\right) \\
& + \frac{D_\ell^{\omega_1^{\beta_\mathrm{d}}\times \omega_1^{\beta_\mathrm{s}}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \ln\left(\frac{\nu_j}{\nu_0}\right) \ln\left(\frac{\nu_i}{\nu_0}\right) \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_0) \right) \\
& + \frac{D_\ell^{\omega_1^{T_\mathrm{d}}\times \omega_1^{\beta_\mathrm{s}}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_j}{\nu_0}\right) (\Theta_j - \Theta_0) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_i}{\nu_0}\right) (\Theta_j - \Theta_0) \right) \\
& + \frac{D_\ell^{A_\mathrm{d}\times \omega_1^{\beta_\mathrm{s}}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_j}{\nu_0}\right) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_i}{\nu_0}\right) \right) \\
& + \frac{D_\ell^{\omega_1^{\beta_\mathrm{d}}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_i}{\nu_0}\right) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) \ln\left(\frac{\nu_j}{\nu_0}\right) \right) \\
& + \frac{D_\ell^{\omega_1^{T_\mathrm{d}}\times A_\mathrm{s}}}{I^\mathrm{s}_{\nu_0}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_0}(\beta_\mathrm{d},T_\mathrm{d})} \left(I^\mathrm{s}_{\nu_j}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_i}(\beta_\mathrm{d},T_\mathrm{d}) (\Theta_i - \Theta_0) + I^\mathrm{s}_{\nu_i}(\beta_\mathrm{s}) I^\mathrm{d}_{\nu_j}(\beta_\mathrm{d},T_\mathrm{d}) (\Theta_j - \Theta_0) \right) \\
& + \cdots
\end{aligned}
\tag{60}
$$

F Parameter Fitting of Foregrounds for the s0d0 Model

In this section, we provide information on the adjustments of the various parameters of the foregrounds for the s0d0 model.

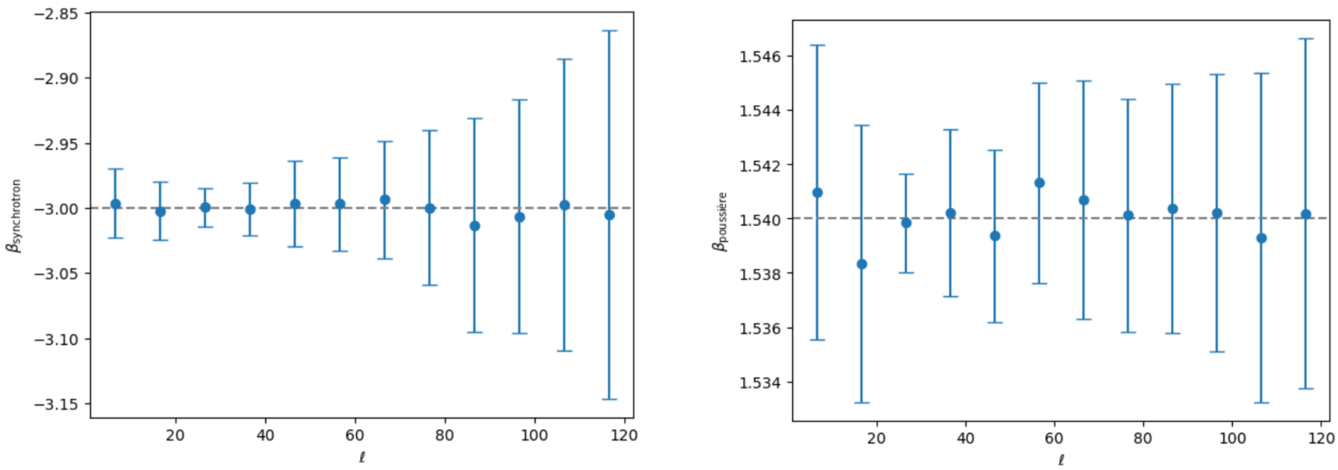


Figure 12: Left: Adjustment of the spectral index of synchrotron, Right: Adjustment of the spectral index of dust; Median and absolute deviation from the median of the estimated parameters. The gray dashed lines correspond to the theoretical value.

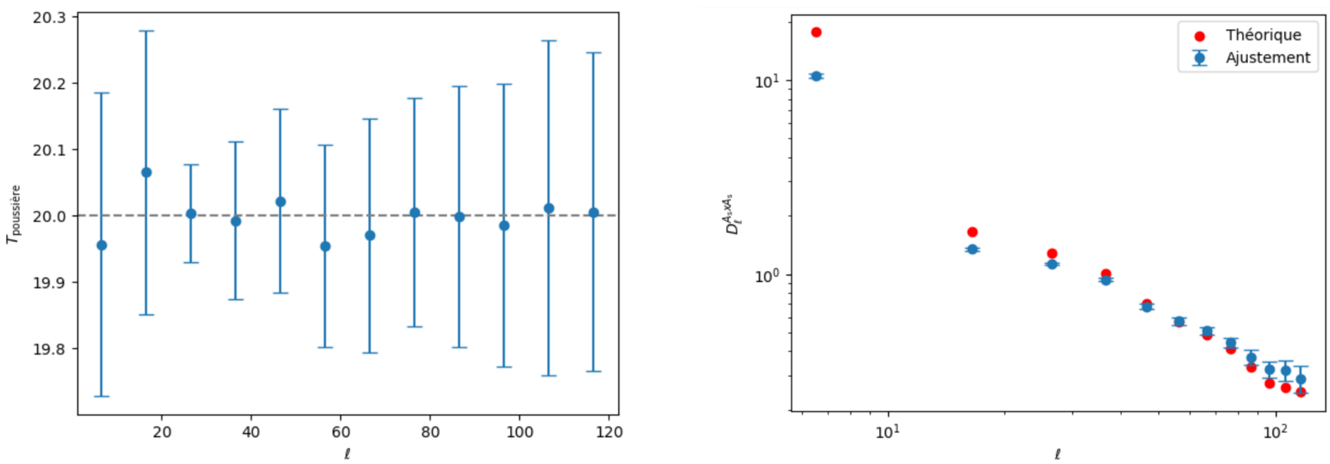


Figure 13: Left: Adjustment of the dust temperature. The gray dashed lines correspond to the theoretical value, Right: Adjustment of the amplitude of the synchrotron, in red the theoretical, in blue the adjustment; Median and absolute deviation from the median of the estimated parameters.

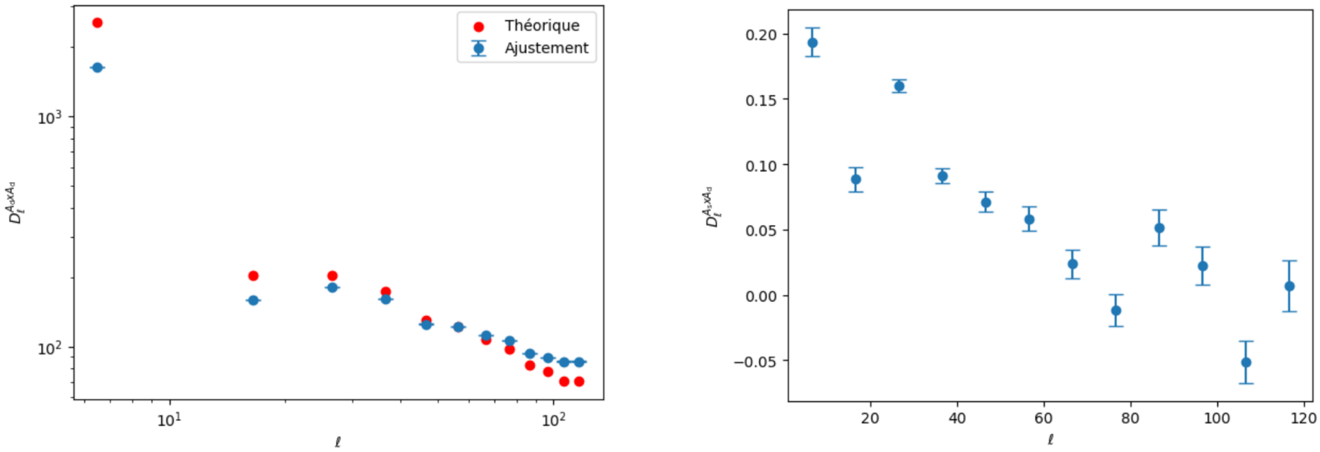
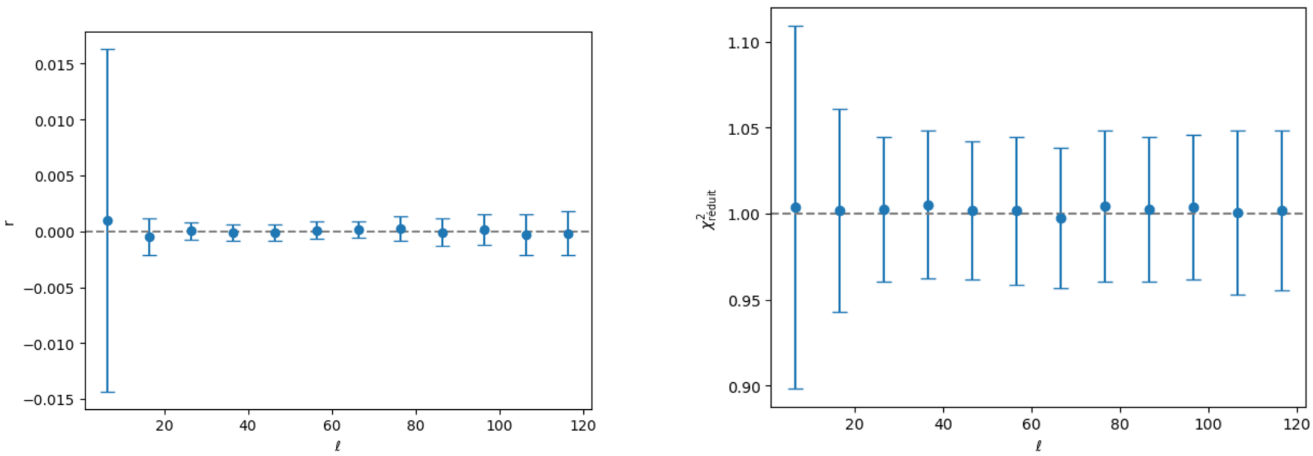


Figure 14: Left: Adjustment of the amplitude of the synchrotron, in red the theoretical, in blue the adjustment; Right: Adjustment of the cross amplitude of synchrotron with dust; Median and absolute deviation from the median of the estimated parameters.



(a) (b)

Figure 15: Left: Adjustment of r. The gray dashed lines correspond to the theoretical value, Right: Reduced χ^2 of the adjustments. The dashed lines correspond to a good fit for $\chi^2_{\text{reduced}} = 1$; Median and absolute deviation from the median of the estimated parameters.

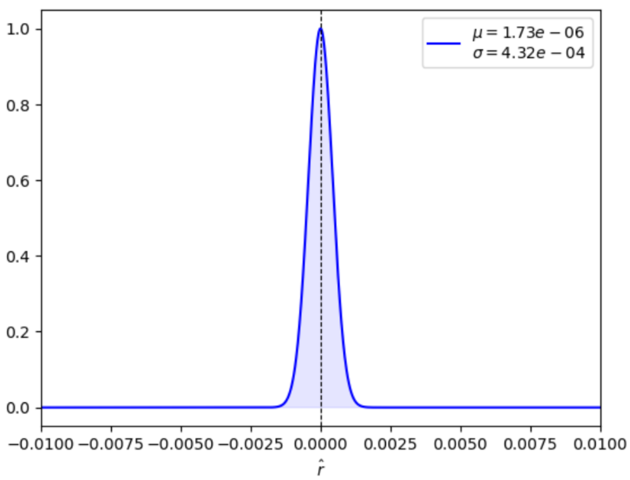


Figure 16: Final result of the estimation of r. The dashed lines correspond to the input value of the simulation $r = 0$.

G Summary of Results

We present the results of the fittings in the form $\hat{r} = \bar{r} \pm \sigma(r)$, with the model used in columns and the order of the moment expansion in rows.

	s0	s1	s7
o(0)	$(3.5 \pm 38.1) \cdot$ e − 5	$(1.65 \pm 3.90) \cdot$ e − 4	$(2.80 \pm 3.87) \cdot$ e − 4
o(β_s)	$(1.38 \pm 4.46) \cdot$ e − 4	$(2.01 \pm 4.54) \cdot$ e − 4	$(8.46 \pm 4.38) \cdot$ e − 4
o(β_s^2)	$(0.06 \pm 9.96) \cdot$ e − 4	$(0.12 \pm 1.04) \cdot$ e − 3	$(0.07 \pm 1.01) \cdot$ e − 3
o(β_s^3)	$(0.58 \pm 1.68) \cdot$ e − 3	$(0.23 \pm 1.59) \cdot$ e − 3	$(0.47 \pm 1.68) \cdot$ e − 3
o(β_d)	✗	✗	✗
o(T_d)	✗	✗	✗
o($\beta_d - T_d$)	✗	✗	✗
o($\beta_d - \beta_d - T_d$)	✗	✗	✗

	d1	d0s0	d0s1
o(0)	$(94.5 \pm 5.3) \cdot$ e − 4	$(0.02 \pm 4.32) \cdot$ e − 4	$(0.84 \pm 4.36) \cdot$ e − 4
o(β_s)	✗	$(0.44 \pm 5.22) \cdot$ e − 4	$(0.68 \pm 4.49) \cdot$ e − 4
o(β_s^2)	✗	$(0.16 \pm 2.74) \cdot$ e − 3	$(-0.18 \pm 2.91) \cdot$ e − 3
o(β_s^3)	✗	$(0.86 \pm 4.33) \cdot$ e − 3	✗
o(β_d)	✗	$(0.45 \pm 5.37) \cdot$ e − 4	✗
o(T_d)	✗	$(0.02 \pm 4.32) \cdot$ e − 4	✗
o($\beta_d - T_d$)	$(-0.39 \pm 1.78) \cdot$ e − 3	$(0.35 \pm 5.28) \cdot$ e − 4	✗
o($\beta_d - \beta_d - T_d$)	✗	$(2.70 \pm 9.07) \cdot$ e − 4	✗

	d0s7	d1s0	d1s1
o(0)	$(3.36 \pm 4.39) \cdot$ e − 4	$(128 \pm 6) \cdot$ e − 4	$(131 \pm 6) \cdot$ e − 4
o(β_s)	$(0.52 \pm 5.00) \cdot$ e − 4	✗	$(154 \pm 7) \cdot$ e − 4
o(β_s^2)	$(-0.01 \pm 2.83) \cdot$ e − 3	✗	✗
o(β_s^3)	✗	✗	✗
o(β_d)	✗	$(27.3 \pm 8.9) \cdot$ e − 4	$(33.0 \pm 8.8) \cdot$ e − 4
o(T_d)	✗	$(81.0 \pm 6.6) \cdot$ e − 4	$(82.7 \pm 6.6) \cdot$ e − 4
o($\beta_d - T_d$)	✗	$(-0.43 \pm 2.28) \cdot$ e − 3	$(0.16 \pm 2.30) \cdot$ e − 3
o($\beta_d - \beta_d - T_d$)	✗	✗	$(0.10 \pm 3.99) \cdot$ e − 3

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